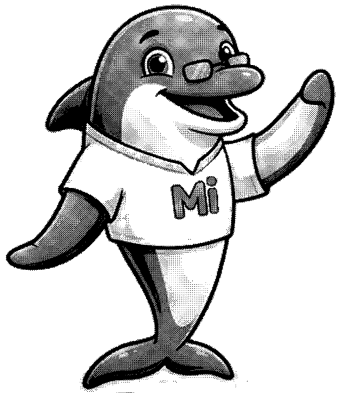
Math Inquirers

# Extension Workbook

## with answers

## Level 1

Self-study | Ages 6-8




My name _____

# About this book

**Math Inquirers Extension Workbook, Level 1**
Self-study practice with answers.


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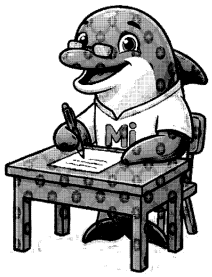

## Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

## Contact

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# Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You do not need a teacher.

**For you**

- ● **Read Learn it. Then do Start here.**
- ◆ **Stuck? Read the hint. It is upside down!**
- ✓ **Do Check yourself. Then look at the answers.**

**For the grown-up**

- Practice for the same ideas as the Holiday Program (our school-holiday maths course), with new numbers and stories. You do not need to have joined it: every unit starts with a worked example.
- 3 short sessions a week, 20 to 25 minutes each. **Go further** is a weekend page: a big challenge for another day (see the pace plan).
- Let your child try before looking at the answers.
- Answers are at the back, from page 79, with a short reason for each. Read the reason together.
- Check yourself score 2 or less? Repeat Learn it and one Start here page.
- Printing the soft copy? Print at 100 percent (actual size) so the rulers are true size.

**The four labels**

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

**Tip:** use a pencil. Mistakes help you learn. Read each question twice before you write.

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| **2** | **Make 10 and 20**<br>Number bonds · follows Holiday Program Lesson 2 · answers on page 81 | **page 13** |
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# My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

| Week | Unit | Session 1<br>Learn it, Start here | Session 2<br>Start here | Session 3<br>Think harder, Check yourself | Weekend page<br>Go further | Done |
|---|---|---|---|---|---|---|
| 1 | **Hops and rules** | 6-7 | 8-9 | 10 and 12 | 11 | ☐ |
| 2 | **Make 10 and 20** | 13-14 | 15-16 | 17 and 19 | 18 | ☐ |
| 3 | **Read and draw graphs** | 20-21 | 22-23 | 24 and 26 | 25 | ☐ |
| 4 | **Patterns that go on** | 27-28 | 29-30 | 31 and 33 | 32 | ☐ |
| 5 | **Back to the start** | 34-35 | 36-37 | 38 and 40 | 39 | ☐ |
| 6 | **Pay the exact amount** | 41-42 | 43-44 | 45 and 47 | 46 | ☐ |
| 7 | **Shape detective** | 48-49 | 50-51 | 52 and 54 | 53 | ☐ |
| 8 | **Clues and sorting** | 55-56 | 57-58 | 59 and 61 | 60 | ☐ |
| 9 | **How long is it?** | 62-63 | 64-65 | 66 and 68 | 67 | ☐ |
| 10 | **Mix and solve** | 69-70 | 71-72 | 73 and 75 | 74 | ☐ |
| 11 | **Catch-up week** | Repeat any Check yourself where your score was 2 or less. | | | | ☐ |

### Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

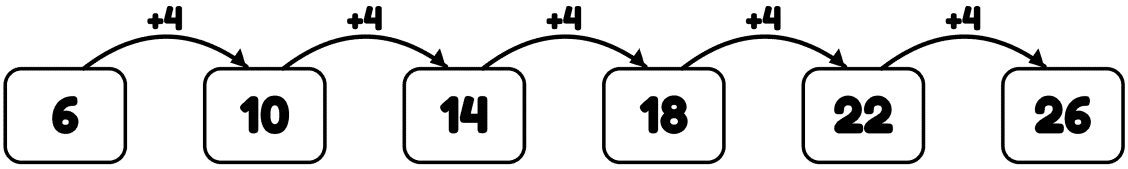
# Learn it: Number patterns

**My goal. I can find the rule of a number pattern and use it to write the next numbers.**

A number pattern follows a rule. The rule tells you how to get from one number to the next. A frog hopping along a number strip makes a pattern.

## Worked example

**Kumar's frog starts on 6. It hops 4 each time. Write the first 6 numbers.**




1. Start on 6. Write 6.
2. Hop 4. 6 and 4 more is 10.
3. Hop 4 again. 10 and 4 more is 14.
4. Keep going: 18, 22, 26. Each number is 4 more than the number before.

**Answer: 6, 10, 14, 18, 22, 26. The rule is add 4.**

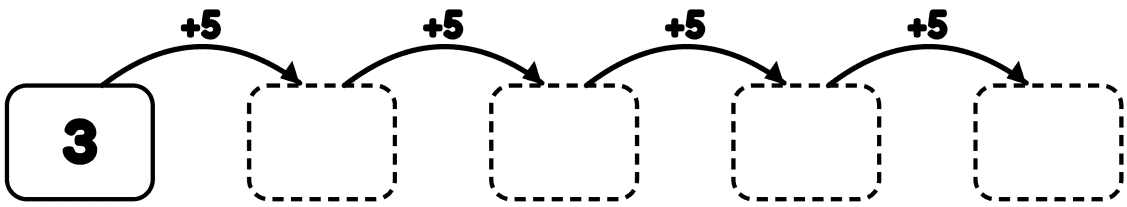
## How to find a rule

1. Look at the first 2 numbers. How far is it from one to the other?
2. Check the next 2 numbers. Is the hop the same?
3. Going up? The rule is add. Going down? The rule is take away.
4. Use the rule to write the next numbers.

## Your turn

Answer at the back of the book.

Try **A frog starts on 3 and hops 5 each time. Write the missing numbers in the tiles.**

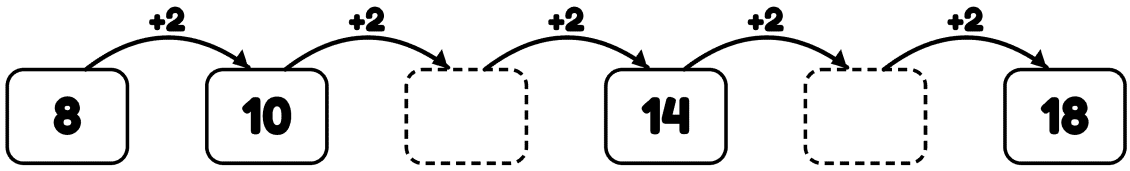


Name _____

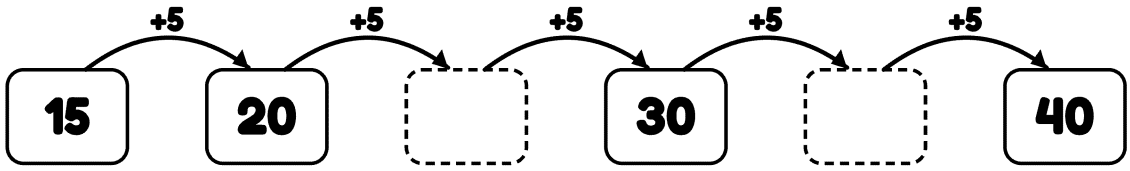
## Start here

Do these first. Take your time.

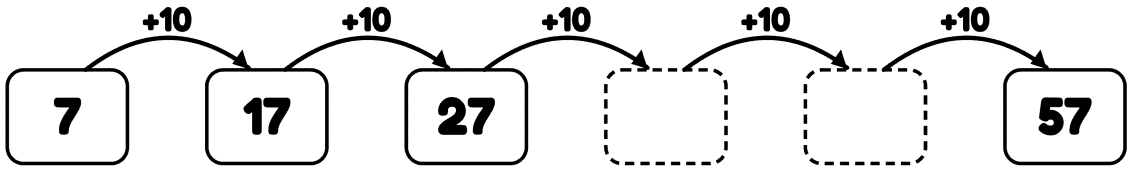
**1** Count in 2s. Write the missing numbers in the tiles.




**2** Count in 5s. Write the missing numbers in the tiles.




**3** Count in 10s. Write the missing numbers in the tiles.




**4** Fill in the blanks.

(a) 5, 9, 13, 17, ☐, ☐

(b) The rule is add ☐.

**5** Fill in the blanks.

(a) 21, 18, 15, 12, ☐, ☐

(b) The rule is take away ☐.

## Start here

Keep going. Check your work.

**6** **Circle the rule for this pattern: 7, 11, 15, 19.**

add 3 add 4 add 5

**7** **Look at the pattern: 30, 26, 22, 18, ...**

(a) Do the numbers go up or down? Up Down

(b) The rule is take away ☐.

(c) The next number is ☐.

**8** **Siti puts 3 stickers in box 1, 6 stickers in box 2 and 9 stickers in box 3. She keeps going.**

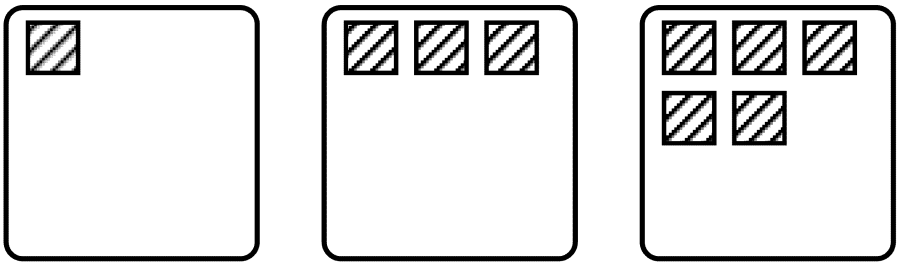
(a) How many stickers go in box 5? ☐ stickers

(b) How many stickers go in box 6? ☐ stickers

Write the numbers: _____

**Hint.** Compare box 1 and box 2. How many more stickers?

**9** **Ben builds steps with blocks.**



Step 1 Step 2 Step 3

(a) How many blocks are in Step 4? ☐ blocks

(b) Each step has ☐ more blocks than the step before.

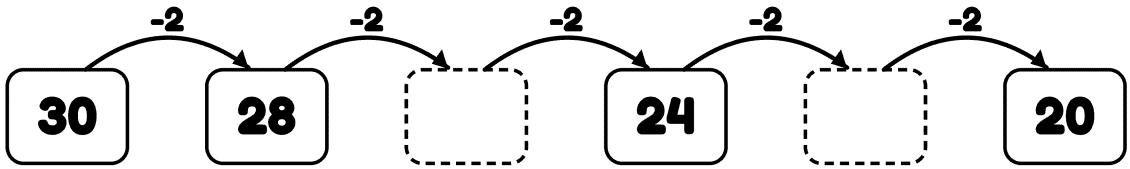
## Start here

Nearly there. You can do it.

**10** **One number is wrong. Circle it. Then write the right number.**
**9, 12, 15, 17, 21, 24**

**The right number is** ☐ **.**

**11** **Count back in 2s. Write the missing numbers in the tiles.**



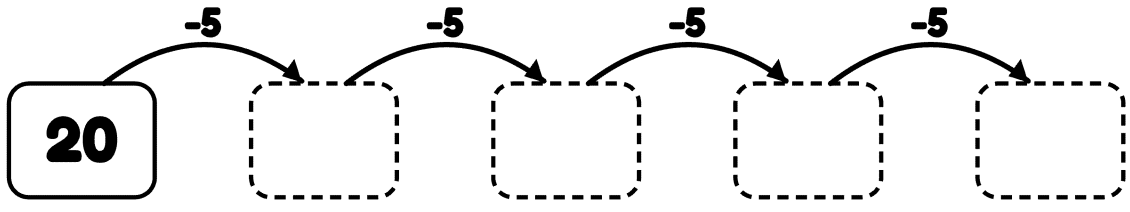

**12** **Wei puts 5¢ in his jar on Monday, 10¢ on Tuesday and 15¢ on Wednesday.**
**He keeps going. How much does he put in on Friday?** ☐ **¢**

Write the days and amounts: _____

**13** **Mei starts on 3 and hops 4 each time. Circle every number she lands on.**

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |

**14** **Kumar starts on 20 and counts back in 5s. Write the missing numbers in the tiles.**



## Think harder

Hints are upside down. Try first!

**1** **Siti starts on 2 and hops 5 each time.**

**(a)** Will she land on 37? Yes No

**(b)** Will she land on 40? Yes No

Write the numbers she lands on: _____

_____

**Hint.** Hop from 2. Write each number you land on.

**2** **Ali counts in 4s from 4 up to 50. Mei counts in 10s from 10 up to 50.**

**(a)** Which numbers do they both say? _____

**(b)** How far apart are the numbers they both say? ☐

Ali: _____

Mei: _____

**Hint.** Write Ali's numbers and Mei's numbers. Look for matches.

**3** **A pattern adds the same number each time. Its 2nd number is 10. Its 5th number is 25.**

**(a)** How many hops from the 2nd number to the 5th number? ☐

**(b)** The rule is add ☐.

**(c)** The 1st number is ☐.

Hops: _____

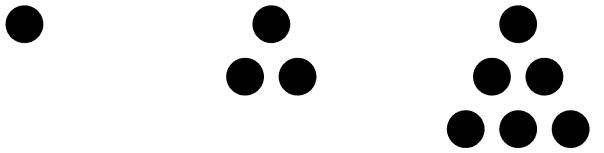
**Hint.** Draw the hops from the 2nd number to the 5th. How far in all?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 1: Triangle numbers

## 1 Look at the triangles made of dots. Triangle 1 has 1 dot, Triangle 2 has 3 dots and Triangle 3 has 6 dots. The pattern keeps going.



Triangle 1 Triangle 2 Triangle 3

(a) Triangle 4 has ☐ dots.

(b) Triangle 5 has ☐ dots.

(c) Triangle 7 has ☐ dots.

**Hint.** How many new dots does each triangle add? Is it the same?

## 2 Mei starts on 1 and adds 3 each time. Ali starts on 2 and adds 5 each time. Both stop at 40. Write the numbers that Mei and Ali both say.

Mei: _____

Ali: _____

Both: _____

**Hint.** List each pattern up to 40. Which numbers are on both lists?

## 3 A pattern starts on 2 and adds the same whole number each time. The pattern says 20. Which numbers can it add? Find them all.

I tried: _____

_____

**Hint.** How far is it from 2 to 20? Try different hops.

## ✔ Check yourself

No hints. Try each one on your own.

**1** 14, 18, 22, 26, ☐, ☐

Working: _____

**2** 40, 35, 30, 25, ☐, ☐. The rule is take away ☐.

Working: _____

**3** Is 30 in this pattern: 6, 10, 14, 18, ...? Yes No

Working: _____

**4** One number is wrong: 11, 16, 21, 27, 31, 36. The right number is ☐.

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

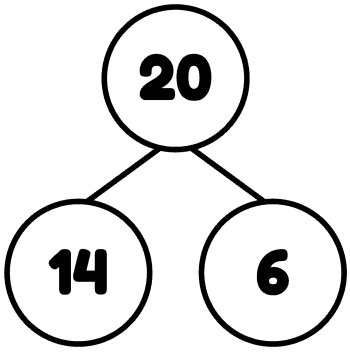
# Learn it: Number bonds

**My goal.** I can find the missing partner in a bond to 10 and to 20. I can tell when the 2 sides are equal.

A number bond is 2 numbers that make a whole. 6 and 4 make 10. So 6 + 4 = 10 and 10 = 6 + 4. Bonds to 10 help us find bonds to 20.

## Worked example

**Aisha has 20 beads. 14 beads are blue and the rest are white. How many beads are white?**




**20 = 14 + 6**

1. The whole is 20. One part is 14. We want the other part.
2. 14 is 10 and 4 more. We need 4 and some more to make 10: 4 + 6 = 10.
3. So 14 + 6 = 20. The other part is 6.
4. Check: 14 + 6 = 20.

**Answer: 6 beads are white. 20 = 14 + 6.**

## How to find the missing partner

1. Find the whole (the greatest number).
2. Look at the ones digit of the part you know. What makes 10 with it?
3. Add the partners. Do they make the whole?

## Your turn

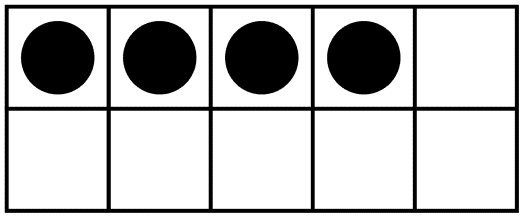
Answer at the back of the book.

Try **Fill in the blank. 20 = 17 +** ☐

## Start here

Do these first. Take your time.

**1** **Look at the ten-frame.**



(a) How many more counters fill the frame? ☐ counters

(b) 10 = 4 + ☐

**2** **Fill in the missing numbers.**

(a) 10 = 8 + ☐

(b) 10 = ☐ + 3

(c) 10 = 5 + ☐

(d) 10 = 1 + ☐

**3** **Kumar has 10 marbles. 6 marbles are in his pocket. The rest are in a bag.**

**How many marbles are in the bag?** ☐ **marbles**

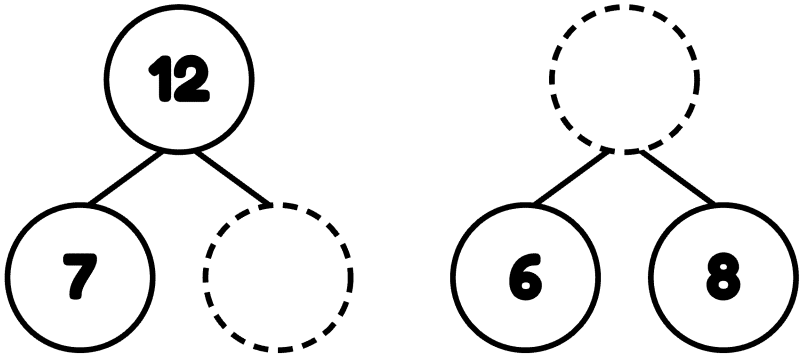
**4** **Write the partner that makes 10.**

**1 and** ☐ **3 and** ☐ **6 and** ☐ **8 and** ☐

## Start here

Keep going. Check your work.

**5** **Write the missing number in each bond.**




**6** **Fill in the missing numbers.**

(a) 20 = 15 + ☐

(b) 20 = ☐ + 14

(c) 20 = 18 + ☐

(d) 20 = 11 + ☐

---

**Hint.** Look at the ones digit. What makes 10 with it?

**7** **7 + 3 = 10. Use it to fill in the blanks.**

17 + ☐ = 20 27 + ☐ = 30 47 + ☐ = 50

**8** **Mei has a shelf for 20 books. There are 16 books on the shelf. How many more books fit?** ☐ **books**

## Start here

Nearly there. You can do it.

**9** **Fill in the missing number. Both sides must be equal.**

(a) 6 + 5 = 7 + ☐

(b) 8 + 2 = 5 + ☐

(c) 9 + 4 = ☐ + 5

**10** **Look at the pairs that make 9.**

(a) 1 + 8 = 9  2 + ☐ = 9  3 + ☐ = 9  4 + ☐ = 9

(b) The first numbers go up by 1. The second numbers go up down by 1.

**11** **Circle True or False.**

(a) 4 + 6 = 10 True False

(b) 8 + 3 = 10 True False

(c) 13 + 7 = 20 True False

(d) 15 + 6 = 20 True False

**12** **There are 20 children in a club. 7 children are girls. The rest are boys. How many boys are in the club?** ☐ **boys**

**13** **Kumar has 8 stickers.**

(a) He wants 10 stickers. How many more does he need? ☐ stickers

(b) He wants 20 stickers. How many more does he need? ☐ stickers

## Think harder

Hints are upside down. Try first!

**1** **Ravi puts 14 marbles into 2 bags. The first bag has fewer marbles than the second bag. Each bag has at least 1 marble. Write every way in the table.**

How many ways are there? ☐

| First bag | Second bag |
|---|---|
| | |
| | |
| | |
| | |
| | |
| | |

**Hint.** Try 1 marble in the first bag. Then 2. Where do you stop?

**2** **Each shape stands for a number. The same shape is always the same number.**

**(a)** ★ + ★ + 5 = 17. ★ = ☐

**(b)** ● + ● + ● = 21. ● = ☐

**Hint.** Cover the 5. What do the 2 stars make together?

**3** **Three different numbers from 1 to 6 add up to 9. Find every set of three.**

☐ + ☐ + ☐ = 9

☐ + ☐ + ☐ = 9

☐ + ☐ + ☐ = 9

**Hint.** Write each set from smallest to greatest. Start with 1.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 1: Making two amounts equal

**1** **Ali has 16 stickers. Mei has 8 stickers. Ali gives some of his stickers to Mei. Now each child has an equal number of stickers.**

(a) How many stickers does Ali give Mei? ☐ stickers

(b) How many stickers does each child have now? ☐ stickers

**Hint.** How many stickers are there in all? What would be fair?

**2** **Mei puts 17 sweets in 2 jars. Jar A has 5 more sweets than Jar B.**

(a) Jar A has ☐ sweets.

(b) Jar B has ☐ sweets.

**Hint.** Imagine Jar A without its 5 extra sweets.

**3** **I am a number. When I add 7, the answer ends in 0 and is between 15 and 45. What can I be? Find them all.**

The answer can be: _____

So I can be: _____

**Hint.** Which numbers between 15 and 45 end in 0?

**4** **Pick 2 different numbers from 1, 2, 3, 4, 5 and add them.**

(a) Write every total you can make. Begin with the smallest.

_____

(b) Which totals can you make in only one way?

_____

My pairs: _____

_____

**Hint.** List the pairs in order. Start with 1 + 2.

Name _____

## ✔ Check yourself

No hints. Try each one on your own.

**1** $10 = 2 + \square$

Working: _____

**2** $20 = 6 + \square$

Working: _____

**3** $7 + 5 = 9 + \square$

Working: _____

**4** There are 20 flowers. 11 are red. The rest are yellow. How many flowers are yellow? ☐ flowers

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# Learn it: Picture graphs

**My goal. I can read a picture graph and use it to add and compare.**

A picture graph uses pictures to show how many. The key tells what 1 picture means.

## Worked example

**The graph shows books borrowed in 1 week. How many more on Tuesday than on Thursday?**

**Books borrowed**

| | |
|---|---|
| Monday | 📖 📖 📖 📖 |
| Tuesday | 📖 📖 📖 📖 📖 📖 📖 |
| Wednesday | 📖 📖 📖 📖 📖 |
| Thursday | 📖 📖 |

**Key: 📖 = 1 book**

1. Read the key. 1 picture is 1 book.
2. Tuesday has 7 books. Thursday has 2 books.
3. "How many more" asks for the gap. Subtract: 7 − 2 = 5. Check: 2 + 5 = 7.

**Answer: 5 more books are borrowed on Tuesday. 7 − 2 = 5.**

## How to read a picture graph

1. Read the title and the key.
2. Count the pictures in each row you need.
3. Add for "in all". Subtract for "how many more".

## Your turn

Answer at the back of the book.

**Try** How many more shells does Ben have than Kumar? ☐ shells

| | |
|---|---|
| Ben | ☆ ☆ ☆ ☆ ☆ |
| Kumar | ☆ ☆ |

**Key: ☆ = 1 shell**

## Start here

Do these first. Take your time.

**1** **Look at the graph. Fill in the blanks.**

**Games at recess**

| | |
|---|---|
| **Skipping** | ☺ ☺ ☺ ☺ ☺ ☺ ☺ |
| **Tag** | ☺ ☺ ☺ ☺ ☺ ☺ ☺ |
| **Catch** | ☺ ☺ ☺ ☺ ☺ |
| **Hopscotch** | ☺ ☺ ☺ ☺ |
| **Marbles** | ☺ ☺ ☺ |

**Key:** ☺ **= 1 child**

(a) ☐ children chose catch.

(b) Which game did the fewest children choose?

_____

(c) Which 2 games got the same number of votes?

_____

**2** **Look at the graph in Question 1.**

(a) How many more children chose skipping than hopscotch? ☐ children

(b) How many children are there altogether? ☐ children

**3** **Look at the graph in Question 1. Circle True or False.**

(a) More children chose catch than hopscotch. True False

(b) Marbles got 4 votes. True False

**4** **30 children vote for a class trip. Zoo gets 9 votes, museum 6 and beach 8. The rest vote for the farm.**

(a) How many children chose the farm? ☐ children

(b) Which place got the most votes? _____

## Start here

Keep going. Check your work.

**5** Look at the graph. Fill in the blanks.

**Stickers this week**

| | |
|---|---|
| Ali | ☆☆☆☆☆☆ |
| Mei | ☆☆☆☆☆☆☆☆ |
| Ben | ☆☆☆ |
| Siti | ☆☆☆☆☆ |

Key: ☆ = 1 sticker

(a) Who has the most stickers? _____

(b) Ben has ☐ stickers.

(c) Mei has ☐ more stickers than Siti.

**6** Look at the graph in Question 5.

(a) How many stickers do Ali and Ben have in all? ☐ stickers

(b) Ali gives 2 stickers to Ben. How many stickers does Ben have now?

☐ stickers

**7** Draw 1 face for each child. 7 children chose water, 4 chose milk and 2 chose tea. How many faces did you draw in all? ☐ faces

**Drinks at the party**

| | |
|---|---|
| Water | ☐☐☐☐☐☐☐☐ |
| Milk | ☐☐☐☐☐☐☐☐ |
| Tea | ☐☐☐☐☐☐☐☐ |

Key: ☺ = 1 child

## Start here

Nearly there. You can do it.

**8** **Look at the graph. Fill in the blanks.**

**Cookies sold at the fair**

| | |
|---|---|
| Monday | ● ● ● ● |
| Tuesday | ● ● ● ● ● ● |
| Wednesday | ● ● ● ● ● ● ● ● |
| Thursday | ● ● ● ● ● |
| Friday | ● ● ● ● ● ● ● |

Key: ● = 1 cookie

(a) On which day were the most cookies sold?

_____

(b) ☐ cookies were sold on Thursday.

(c) How many more cookies were sold on Friday than on Monday?

☐ cookies

**9** **Ali has 5 books. Mei has 3 more books than Ali. Draw Mei's row. How many books does Mei have? ☐ books**

| | |
|---|---|
| Ali | 📖 📖 📖 📖 📖 |
| Mei | ▯ ▯ ▯ ▯ ▯ ▯ ▯ ▯ |
| Ben | 📖 📖 📖 📖 |

Key: 📖 = 1 book

**10** **Ravi makes 12 treats in all. 7 are cookies. The rest are cakes. Draw the cakes. How many cakes does Ravi make? ☐ cakes**

| | |
|---|---|
| Cookies | ● ● ● ● ● ● ● |
| Cakes | ▯ ▯ ▯ ▯ ▯ ▯ ▯ ▯ |

Key: ● = 1 treat

## Think harder

Hints are upside down. Try first!

★ Beyond Level 1: Keys bigger than 1

**1** **Each face stands for 2 children. A half face stands for 1 child.**

**Children in clubs**

| | |
|---|---|
| Drawing | ☺ ☺ ☺ ☺ |
| Singing | ☺ ☺ ☺ |
| Chess | ☺ ☺ ☺ ☺ ☺ ◔ |

**Key:** ☺ = 2 children ◔ = 1 child

(a) How many children chose chess? ☐ children

(b) How many more children chose chess than singing? ☐ children

**Hint.** Count in 2s along each row. What does a half face add?

**2** **Siti asks 5 friends which game they like best. 4 friends choose tag. Is it true that most children in the school like tag best?**

Yes No We cannot tell

**Hint.** Do 5 friends stand for every child in the school?

★ Beyond Level 1: Comparing two keys

**3** **Look at the 2 graphs. Their keys are not the same.**

| | |
|---|---|
| Graph P | ☺ ☺ ☺ ☺ ☺ ☺ |

**Key:** ☺ = 1 child

| | |
|---|---|
| Graph Q | ☺ ☺ |

**Key:** ☺ = 5 children

(a) Graph P shows ☐ children. Graph Q shows ☐ children.

(b) Which graph shows more children? Graph P Graph Q

**Hint.** Read each key first. What does one face mean?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ **Beyond Level 1: Build a graph from clues**

## 1 Ravi, Lily, Jamie and Ray find 24 shells in all. Ravi finds 7. Lily finds 2 fewer than Ravi. Jamie and Ray find the same number.

| | |
|---|---|
| **Ravi** | ☆ ☆ ☆ ☆ ☆ ☆ ☆ |
| **Lily** | ▢ ▢ ▢ ▢ ▢ ▢ ▢ ▢ |
| **Jamie** | ▢ ▢ ▢ ▢ ▢ ▢ ▢ ▢ |
| **Ray** | ▢ ▢ ▢ ▢ ▢ ▢ ▢ ▢ |

**Key:** ☆ **= 1 shell**

(a) Lily finds ☐ shells.

(b) Jamie finds ☐ shells.

**Hint.** Who can you find first? Then what is left?

## 2 A graph has rows for kites, balls and ropes. There are 18 toys. There are 2 more balls than kites. There is 1 more rope than kites.

(a) Kites: ☐

(b) Balls: ☐

(c) Ropes: ☐

**Hint.** Take away the extras first. What is left?

## 3 In Wei's graph, each picture stands for 2 stickers. Wei has 6 pictures. Siti has the same number of stickers as Wei. In Siti's graph, each picture stands for 3 stickers.

(a) How many stickers does Wei have? ☐ stickers

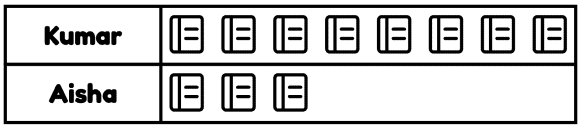
(b) How many pictures does Siti draw? ☐ pictures

**Hint.** How many stickers does Wei have? Then count in 3s.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Use the graph for Questions 1 to 4. Kumar read ☐ books.**




**Key:** 📖 **= 1 book**

**2** **How many more books did Kumar read than Aisha? ☐ books**

Working: _____

**3** **How many books did the 2 children read in all? ☐ books**

Working: _____

**4** **Who read the fewest books?** _____

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

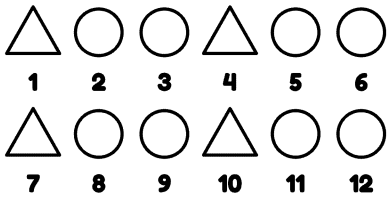
# Learn it: Repeating and growing patterns

**My goal.** I can find the core of a pattern and use it to find a shape far along. I can follow a growing pattern and find the rule.

A repeating pattern has a core that goes round and round. Count in the size of the core to find a far place. A growing pattern gets larger by the same amount each time.

## Worked example 1

**The pattern keeps going. What is the 14th shape?**




1. Find the core. It is triangle, circle, circle: 3 shapes.
2. Count in 3s: 3, 6, 9, 12. Place 12 is the end of a core.
3. Places 13 and 14 start a new core. Place 13 is a triangle and place 14 is a circle.

**Answer: The 14th shape is a circle.**

## Worked example 2

**Ali has 4 marbles on day 1. Each day he has 5 more than the day before. How many marbles on day 5?**

| Day | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Marbles | 4 | 9 | 14 | 19 | 24 |

+5 +5 +5 +5

1. Make a table. Write 4 under day 1.
2. Each day has 5 more. The rule is add 5: 9, 14, 19, 24.
3. Check: 4 hops of 5 is 20. 4 + 20 = 24.

**Answer: On day 5 he has 24 marbles.**

## How to find a far place

1. Find the core and count its shapes.
2. Count in that size. Stop at or before your place.
3. Count on from the last full core.

## Your turn

Answer at the back of the book.

Try

**The pattern keeps going. What is the 10th shape?**

**circle square triangle star**



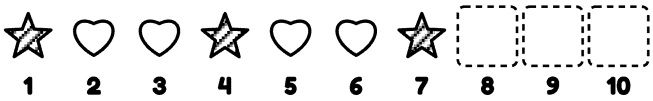
## Start here

Do these first. Take your time.

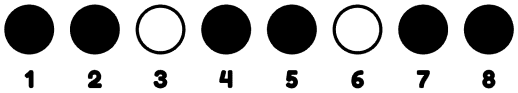
**1** Circle the shape that comes next. triangle circle




**2** Draw the next 3 shapes in the boxes. Then circle the 9th shape.
star heart



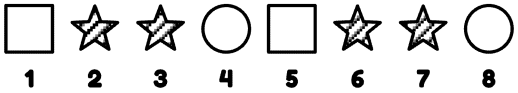

**3** Dark beads and white beads make a pattern. What is the 12th bead?
dark white




**4** Write the next 2 letters. M, N, O, M, N, O, M, _____ _____

**5** Ravi claps and stomps. The pattern is clap, clap, stomp, clap, clap, stomp, and so on. What is the 10th action? clap stomp

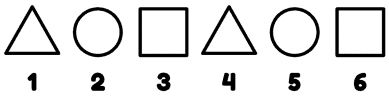
**6** The pattern keeps going. What is the 16th shape?
square star circle



## Start here

Keep going. Check your work.

**7** **Look at the pattern. Circle the right shape.**




(a) The 9th shape is a triangle circle square .

(b) The 12th shape is a triangle circle square .

(c) The 13th shape is a triangle circle square .

**8** **Look at the pattern in Question 7. The circles are in places 2, 5 and 8.**

**Write the next 3 places that have a circle.** ☐ ☐ ☐

**9** **A pattern repeats every 4 shapes. Circle True or False.**

(a) The 8th shape is the same as the 4th shape. True False

(b) The 10th shape is the same as the 4th shape. True False

(c) The 13th shape is the same as the 1st shape. True False

**10** **The pattern keeps going. How many circles are in the first 12 shapes?**

☐ **circles**




**11** **Lily puts 3 beads on a string on day 1. She adds 2 beads each day.**

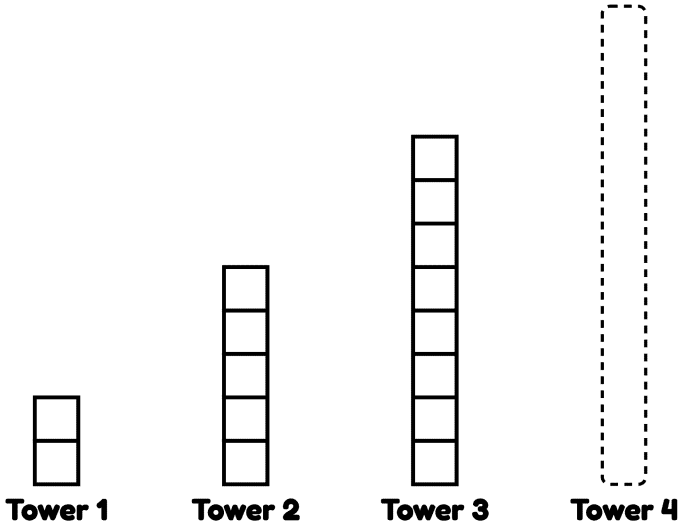
(a) How many beads does she have on day 5? ☐ beads

(b) On which day does she have 15 beads? Day ☐

## Start here

Nearly there. You can do it.

**12** **The towers grow in a pattern. Draw Tower 4. Fill in the blanks.**

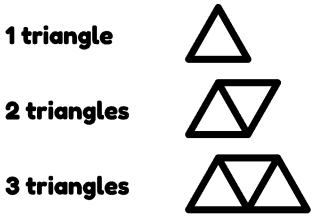



(a) Tower 4 has ☐ blocks.

(b) Tower 6 has ☐ blocks.

(c) The tower with 23 blocks is Tower ☐.

**13** **Aisha makes a row of triangles with straws.**




(a) Count the straws. 2 triangles use ☐ straws.

(b) 3 triangles use ☐ straws.

(c) 4 triangles use ☐ straws.

## Think harder

Hints are upside down. Try first!

**1** **The pattern keeps going.**



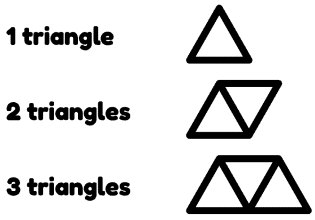

(a) What is the 15th shape? heart star

(b) How many stars are in the first 18 shapes? ☐ stars

(c) In which place is the 5th star? Place ☐

**Hint.** Find the core. How many stars are in one core?

**2** **Siti has 24 straws. She makes a row of triangles.**




(a) How many triangles can she make? ☐ triangles

(b) How many straws are left? ☐ straw

**Hint.** How many straws for 1 triangle? How many for each new one?

**3** **Mei has 5 stickers on day 1, 8 stickers on day 2 and 11 stickers on day 3. The pattern keeps going.**

(a) How many stickers does she have on day 6? ☐ stickers

(b) On which day does she have 32 stickers? Day ☐

**Hint.** Make a table. Write the day and the stickers.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 1: Jumps that grow

**1** **Look at the number pattern: 1, 2, 4, 7, 11. The jumps are not the same.**

(a) Write the next 2 numbers. ☐ ☐

(b) What is the 8th number in the pattern? ☐

**Hint.** Write the jump between each pair of numbers.

★ Beyond Level 1: Two ways to build

**2** **Ravi makes 3 triangles that do not touch. Mei makes a row of 3 triangles that share sides.**

Ravi 

Mei 

(a) Ravi needs ☐ straws for 7 triangles.

(b) Mei needs ☐ straws for 7 triangles.

(c) Mei uses ☐ fewer straws than Ravi for 7 triangles.

**Hint.** How many straws does each new triangle add?

★ Beyond Level 1: Two patterns together

**3** **Mei lays 2 patterns, one above the other. The top pattern is circle, star, circle, star. The bottom pattern is A, B, C, A, B, C.**



| A | B | C | A | B | C | A | B |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

(a) The first place with a star above a C is place ☐.

(b) The next place with a star above a C is place ☐.

**Hint.** Write the places of the stars. Then the places of the Cs.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **The pattern is sun, moon, moon, sun, moon, moon. The 9th one is**

**sun** **moon**

Working: _____

**2** **The letters P, Q, R, S go round and round. The 18th letter is**

**P** **Q** **R** **S**

Working: _____

**3** **Tower 1 has 3 blocks. Each tower adds 4. Tower 5 has ☐ blocks.**

Working: _____

**4** **1 triangle uses 3 straws. Each new one adds 2. 6 triangles use ☐ straws.**

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

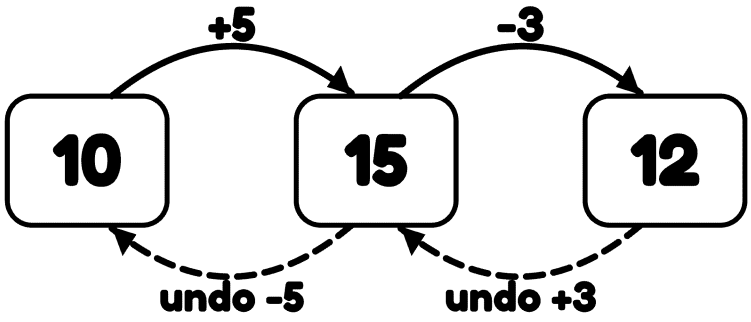
# Learn it: Working backwards

**My goal. I can find the start of a story when I know the end. I go backwards and undo each step.**

In a mystery number story, we know the end. We undo each step.

## Worked example

**Ravi has some stickers. He gets 5 more, then gives 3 to Mei. Now he has 12 stickers. How many stickers did Ravi begin with?**




1. Go backwards. Undo -3 with +3: 12 + 3 = 15.
2. Undo +5 with -5: 15 - 5 = 10.
3. Check going forwards: 10 + 5 = 15 and 15 - 3 = 12.

**Answer: Ravi had 10 stickers at first.**

## How to work backwards

1. Find the end number. This is the number you know.
2. Undo the last step first. Add becomes take away. Take away becomes add.
3. Keep going back until you reach the start.
4. Check by going forwards.

## Your turn

Answer at the back of the book.

Try **Mei buys 6 beads. Now she has 15 beads. How many beads did she begin with? ☐ beads**

## Start here

Do these first. Take your time.

**1** A cup hides some counters. 5 more counters go in. Now there are 13 counters. How many counters were in the cup at first? ☐ counters

**2** Mei picks a secret number. Fill in her secret number each time.

(a) She adds 8 and gets 13. Her number is ☐.

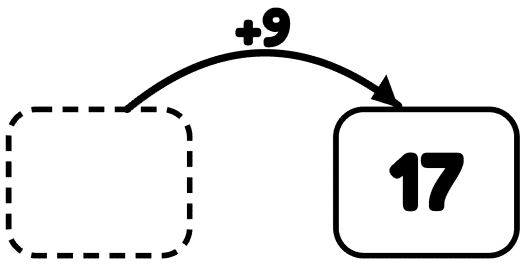
(b) She adds 7 and gets 15. Her number is ☐.

(c) She takes away 5 and gets 9. Her number is ☐.

(d) She takes away 6 and gets 12. Her number is ☐.

**3** Aisha has some biscuits. She eats 4 biscuits. 9 biscuits are left. How many biscuits did she have at first? ☐ biscuits

**4** Write the number in the empty box. Go backwards along the arrow.

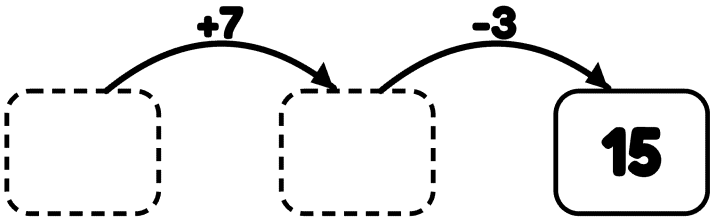



**5** Siti has some pencils. She gives 3 to Ali, then buys 8 more. Now she has 14 pencils. How many pencils did she begin with? ☐ pencils

## Start here

Keep going. Check your work.

**6** **Write the 2 missing numbers in the empty boxes. Go backwards.**




**7** **This machine always adds the same number. Look at the table.**

| IN | 6 | 9 | | 15 |
|---|---|---|---|---|
| OUT | 10 | | 17 | |

(a) The machine adds ☐.

(b) Write the 3 missing numbers in the table.

**8** **Fill in the missing number. Go backwards.**

(a) ☐ + 6 = 15

(b) ☐ - 4 = 9

(c) 18 - ☐ = 11

**9** **A lift goes down 4 floors. Then it goes up 6 floors. It stops on floor 11. On which floor did the lift begin? Floor** ☐

## Start here

Nearly there. You can do it.

**10** **Kumar is 6th in a queue. There are 2 children behind him. How many children are there in the queue altogether? ☐ children**

Draw the queue: _____

**11** **Aisha stands 5th from the front in a line of 12 children. How many children are behind her? ☐ children**

Draw the queue: _____

**12** **Think about more and fewer.**

(a) Ali has 15 stickers. He has 6 more than Siti. Siti has ☐ stickers.

(b) Ben has 9 sweets. He has 6 fewer than Jenny. Jenny has ☐ sweets.

**13** **Check going forwards. Circle True or False.**

(a) Start at 9. Add 4. Take away 6. You get 7. True False

(b) Start at 12. Take away 5. Add 3. You get 9. True False

(c) Start at 8. Add 7. Take away 4. You get 11. True False

**14** **Kumar picks a number and adds 6. He gets 15. Is his number 21?**

(a) Circle Yes or No. Yes No

(b) His number is ☐.

## Think harder

Hints are upside down. Try first!

**1** **Mei has some stickers. She gives 5 to Ali, gets 8 from Ben and gives 4 to Siti. Now she has 13 stickers.**

(a) Before she gave 4 to Siti, she had ☐ stickers.

(b) Before Ben gave her 8, she had ☐ stickers.

(c) At first Mei had ☐ stickers.

**Hint.** Go back one step at a time. Which step is the last?

**2** **A machine adds 9 and then takes away 4. A one-step machine does the same job.**

(a) The one-step machine adds ☐.

(b) IN 11 gives OUT ☐.

(c) OUT 30 comes from IN ☐.

**Hint.** Try IN 10. What is OUT? How far is it from 10?

**3** **Ali gives 4 stickers to Mei. Now each child has 10 stickers.**

(a) Ali had ☐ stickers at first.

(b) Mei had ☐ stickers at first.

Ali: _____

Mei: _____

**Hint.** Who gets 4 back? Who gives 4 back?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 1: Undoing a double

**1** **Aisha picks a number and adds it to itself. This is a double. Find each number.**

(a) 6 + 6 = ☐

(b) Double it, then add 4. She gets 18. Her number is ☐.

(c) Double it, then take away 5. She gets 11. Her number is ☐.

**Hint.** Undo the last step first. Then which double is that?

**2** **Does the order of the steps matter?**

(a) Start at 12. Add 6, then take away 9. You get ☐.

(b) Start at 12. Take away 9, then add 6. You get ☐.

(c) Start at 5. Add 6, then take away 9. You get ☐.

(d) Start at 5. Can you take away 9 first? Yes No

**Hint.** Do each one a step at a time. Check part (d) with care.

**3** **A frog is at the bottom of a well with 12 steps. Each day it climbs 4 steps. Each night it slides back 2 steps. On which day does the frog get out?**

(a) The frog gets out on day ☐.

Day 1: _____

Day 2: _____

Day 3: _____

Day 4: _____

Day 5: _____

**Hint.** Where is the frog after day 1? And after night 1?

## ✔ Check yourself

No hints. Try each one on your own.

**1** Wei picks a number and adds 7. He gets 16. His number is ☐.

Working: _____

**2** Lily gives away 5 stickers, then gets 9. Now she has 17. At first she had ☐ stickers.

Working: _____

**3** A machine takes away 6. OUT is 9. IN is ☐.

Working: _____

**4** Ali picks a number and takes away 7. He gets 8. His number is ☐.

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

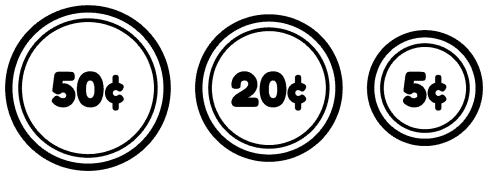
# Learn it: Money

**My goal.** I can count coins and notes and pay an exact amount. I can find all the ways and the fewest coins.

Our coins are 5¢, 10¢, 20¢ and 50¢. A $1 coin is worth 100¢. To pay an amount, start with the largest coin that fits.

## Worked example

**Ben wants to pay exactly 75¢. He has 5¢, 10¢, 20¢ and 50¢ coins. Which coins use the fewest coins?**




1. Take the largest coin that fits: 50¢. 75¢ - 50¢ = 25¢ is left.
2. The largest coin that fits 25¢ is 20¢. 25¢ - 20¢ = 5¢ is left.
3. Take a 5¢ coin. Nothing is left.
4. Check: 50¢ + 20¢ + 5¢ = 75¢. That is 3 coins.

**Answer: 50¢, 20¢ and 5¢. That is 3 coins.**

## How to pay an exact amount

1. Take the largest coin that is not too much.
2. Subtract it from the amount. What is left?
3. Repeat with what is left. Stop at 0.
4. Add your coins to check.

## Your turn

Answer at the back of the book.

Try **Pay exactly 80¢ with 3 coins. Use 5¢, 10¢, 20¢ or 50¢ coins. Write the coins.**

☐ ¢ + ☐ ¢ + ☐ ¢

## Start here

Do these first. Take your time.

**1** Count the money. Write the total. ☐ ¢




**2** Count the money. Write the total. $ ☐




**3** Fill in how many coins.

(a) 50¢ can be changed for ☐ coins of 10¢.

(b) 20¢ can be changed for ☐ coins of 5¢.

(c) $1 can be changed for ☐ coins of 20¢.

(d) $1 can be changed for ☐ coins of 50¢.

**4** Pay exactly with 2 coins. Use 5¢, 10¢, 20¢ or 50¢ coins. Write the 2 coins.

(a) 25¢ = ☐ ¢ + ☐ ¢

(b) 60¢ = ☐ ¢ + ☐ ¢

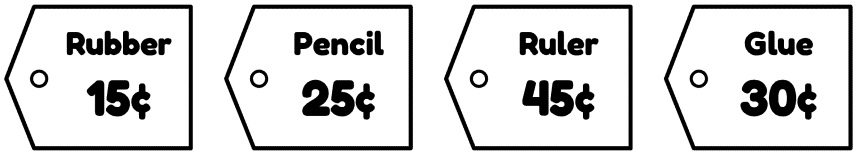
(c) 30¢ = ☐ ¢ + ☐ ¢

**5** Two of me make 40¢. Which coin am I? 5¢ 10¢ 20¢ 50¢

## Start here

Keep going. Check your work.

**6** Look at the tags.



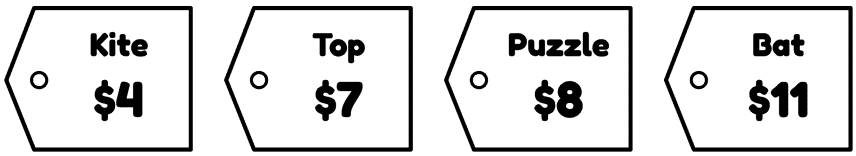

(a) A rubber and a pencil cost ☐ ¢ altogether.

(b) Which costs more, the ruler or the glue? ruler glue

(c) The ruler costs ☐ ¢ more than the rubber.

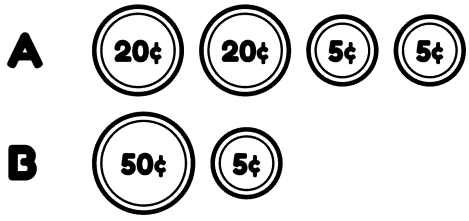
**7** Ben pays 50¢ for a 30¢ sticker. How much change does he get? ☐ ¢

**8** Ali buys 2 toys for exactly $12. Which 2 toys? Circle them.




**9** Lily has $8. A book costs $13. How much more does she need? $ ☐

**10** Look at the 2 purses.



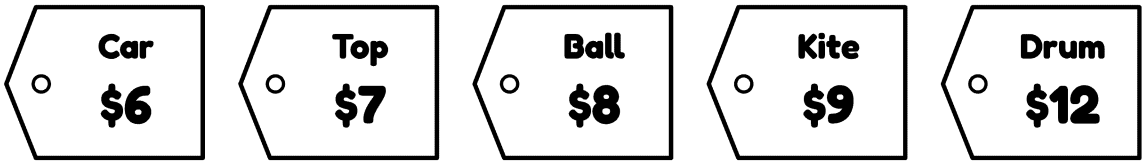

(a) Which purse has more money? A B

(b) It has ☐ ¢ more.

## Start here

Nearly there. You can do it.

**11** **Mei buys exactly 2 toys for $15. There are 2 ways.**




(a) The Car and the Top Ball Kite Drum

(b) The Top and the Car Ball Kite Drum

Try: _____

**12** **Circle True or False.**

(a) Two 20¢ coins and one 10¢ coin make 50¢. True False

(b) Three 10¢ coins make 40¢. True False

(c) A $1 coin is worth ten 10¢ coins. True False

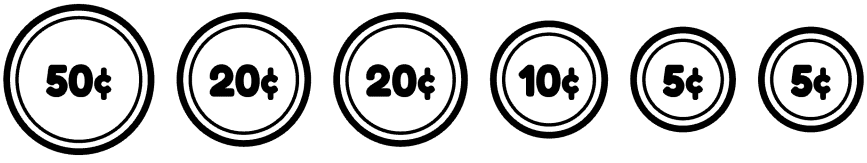
(d) Four 5¢ coins make 25¢. True False

**13** **Pay with the fewest coins. Use 5¢, 10¢, 20¢ or 50¢ coins.**

(a) 40¢ needs ☐ coins.

(b) 90¢ needs ☐ coins.

**14** **Circle the coins that pay exactly 65¢.**



## Think harder

Hints are upside down. Try first!

**1** **Make 65¢ with exactly 4 coins. Use 5¢, 10¢, 20¢ or 50¢ coins. Find every set.**

Set 1: _____

Set 2: _____

**Hint.** Try the 50¢ coin first. What is left to pay?

**2** **Siti has 7 coins. They are 5¢ coins and 10¢ coins. Together they are worth 50¢.**

(a) She has ☐ coins of 5¢.

(b) She has ☐ coins of 10¢.

| 5¢ coins | 10¢ coins | Total |
|---|---|---|
| | | |
| | | |
| | | |

**Hint.** Guess how many are 5¢ coins. Does the total match?

**3** **Make 40¢ with only 20¢ and 10¢ coins. Write every way in the table. Each row is one way.**

(a) Write the ways in the table.

(b) There are ☐ ways.

| 20¢ coins | 10¢ coins |
|---|---|
| | |
| | |
| | |

**Hint.** Start with the most 20¢ coins you can use.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 1: Sharing money equally

## 1 Friends share money equally.

(a) 2 friends share 80¢. Each gets ☐ ¢.

(b) 4 friends share 80¢. Each gets ☐ ¢.

(c) Siti and Wei share 50¢. They have only 20¢ and 10¢ coins. Can each child get the same amount? Yes No

Half of 50¢: _____

**Hint.** What is half of 50¢? Can those coins make it?

## 2 Wei pays with a $50 note for a bag and a hat. The hat costs $18. He gets $12 change.

(a) Wei spends $ ☐ altogether.

(b) The bag costs $ ☐.

Spent: _____

Bag: _____

**Hint.** How much money did Wei really spend?

## 3 Ali pays exactly $38. He has $1 coins and $2, $5 and $10 notes. He uses the fewest notes and coins.

(a) He uses ☐ notes and coins.

(b) They are: _____

Try: _____

**Hint.** Start with the largest note that fits. What is left?

## ✔ Check yourself

No hints. Try each one on your own.

**1** Count the money. Write the total. ☐ ¢




**2** Pay exactly 70¢ with 3 coins. ☐ ¢ + ☐ ¢ + ☐ ¢

Working: _____

**3** Ravi has 90¢. A drink costs 65¢. How much is left? ☐ ¢

Working: _____

**4** Which is more money? 85¢ $1

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

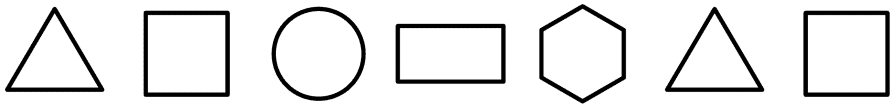
# Learn it: 2D shapes

**My goal. I can name shapes by their sides and corners. I can count shapes, squares and triangles in a picture.**

Shapes have straight sides and corners. A triangle has 3 sides and 3 corners. A circle has none. A square and a rectangle have 4 of each.

## Worked example

**Look at the shapes. How many shapes have 4 corners?**



1. A shape with 4 corners has 4 straight sides.
2. Check each shape. A square has 4 corners. A rectangle has 4 corners.
3. There are 2 squares and 1 rectangle.
4. The other shapes do not have 4 corners.

**Answer: 3 shapes.**

## How to count shapes

1. Pick one kind of shape.
2. Touch each one once. Make a small mark on it.
3. Write down how many.
4. Do the next kind. Add all the counts to check.

## Shape names

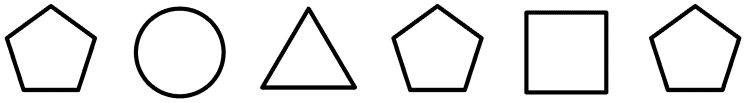


| triangle | square | pentagon | hexagon | octagon |
|---|---|---|---|---|
| 3 sides | 4 sides | 5 sides | 6 sides | 8 sides |

## Your turn

Answer at the back of the book.

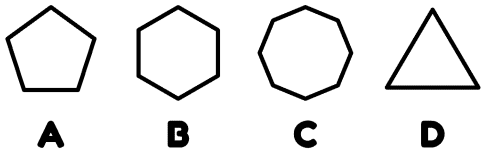
Try **How many pentagons are in the picture?** ☐



## Start here

Do these first. Take your time.

**1** **Look at the 4 shapes. Circle the right name for each shape.**




(a) Shape A is: pentagon hexagon octagon

(b) Shape B is: pentagon hexagon octagon

(c) Shape C is: pentagon hexagon octagon

(d) Shape D is: triangle square circle

**2** **Fill in the blanks.**

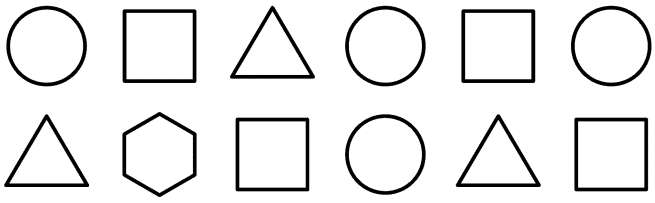
(a) A pentagon has ☐ corners.

(b) A shape with 7 sides has ☐ corners.

(c) A circle has ☐ straight sides.

(d) A shape with 8 corners has ☐ sides.

**3** **Count each kind of shape. Write how many.**



(a) There are ☐ circles.

(b) There are ☐ triangles.

(c) There are ☐ shapes altogether.

## Start here

Keep going. Check your work.

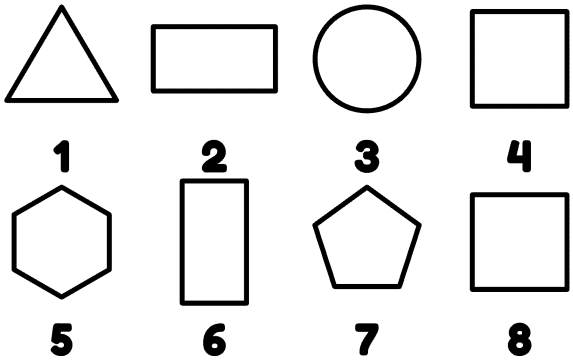
**4** **Read each riddle. Circle the shape.**

(a) I have 5 corners. All my sides are straight.
triangle pentagon hexagon

(b) I have 4 corners. All my sides are the same length.
square triangle circle

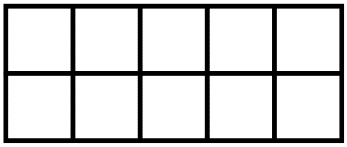
(c) I have 8 straight sides. hexagon octagon pentagon

**5** **Look at the shapes. Which shapes have 4 corners?**

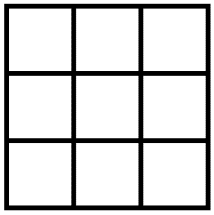



Write their numbers: _____

**6** **Look at the 2 grids. Count the small squares.**



Grid A



Grid B

(a) Grid A has ☐ small squares.

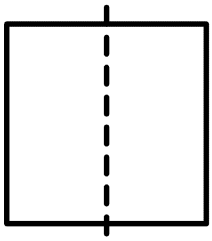
(b) Grid B has ☐ small squares.

(c) Which grid has more small squares? A B

## Start here

Nearly there. You can do it.

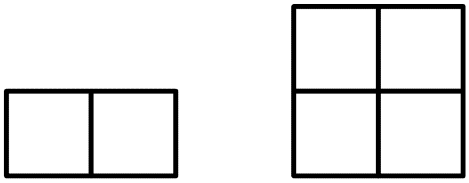
**7** **A square is cut along the dashed line.**



(a) Each piece is a triangle square rectangle

(b) Each piece has ☐ corners.

**8** **Squares join to make Shape A and Shape B.**

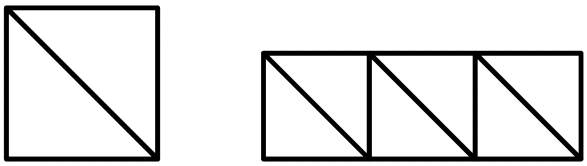


A B

(a) Shape A is a square rectangle triangle

(b) Shape B is a square triangle circle

**9** **Count the triangles. Count the small ones.**



A B

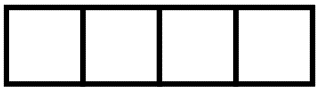
(a) Picture A has ☐ triangles.

(b) Picture B has ☐ triangles.

## Think harder

Hints are upside down. Try first!

**1** **This strip has 4 squares in a row. Count the rectangles in it. A square counts as a rectangle.**



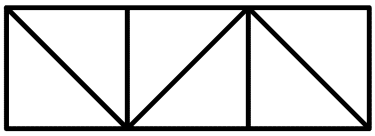
(a) Rectangles that are 1 square long: ☐

(b) Rectangles that are 2 squares long: ☐

(c) All the rectangles in the strip: ☐

**Hint.** Slide a 2 square rectangle along the strip.

**2** **A strip of 3 squares has 3 lines drawn in it. Count the triangles. Small ones first.**



(a) Small triangles: ☐

(b) All the triangles: ☐

**Hint.** Do 2 lines that meet make a larger triangle?

**3** **Add the sides or the corners.**

(a) 2 shapes have 11 straight sides altogether. One is a pentagon. The other is a square hexagon octagon

(b) A triangle, a square and 1 more shape have 15 corners altogether. The last shape has ☐ corners.

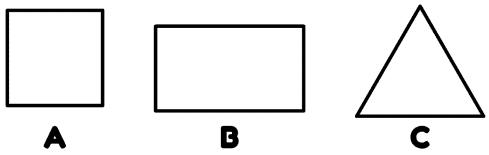
**Hint.** Add the sides you know. How many are left?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 1: Fold lines

## 1 Fold a shape so that both halves match exactly. Each way to fold is a fold line. Draw them.




(a) Square A has ☐ fold lines.

(b) Rectangle B has ☐ fold lines.

(c) Triangle C has 3 sides of the same length. It has ☐ fold lines.

**Hint.** Imagine a mirror on the line. Both sides must match.

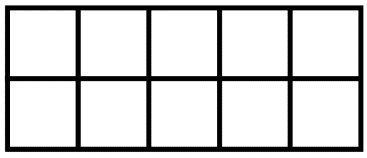
## 2 Jenny uses all her square tiles to make a rectangle. A turned rectangle is the same one. A square counts as a rectangle.

(a) With 12 tiles she can make ☐ different rectangles.

(b) With 9 tiles she can make ☐ different rectangles.

**Hint.** Try rows of 1 tile, then 2, then 3. Does it fit exactly?

## 3 Count the squares in this grid. Squares of every size count.



(a) Squares made of 4 small squares: ☐

(b) All the squares: ☐

**Hint.** Slide a 2 by 2 square along. How many places can it sit?

## ✔ Check yourself

No hints. Try each one on your own.

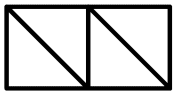
**1** A shape has 5 straight sides. It is a **square** **pentagon** **hexagon**

Working: _____

**2** How many corners do 3 squares have altogether? ☐

Working: _____

**3** How many triangles are in the picture? ☐



**4** A shape has 3 more sides than a triangle. It has ☐ sides.

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# Learn it: Logic and sorting

**My goal.** I can use clues, fill in a logic grid and sort numbers into loops.

## Worked example 1

**Ravi, Jenny and Wei each pick milk, juice or tea. Ravi picks tea. Jenny does not pick milk. Who picks what?**

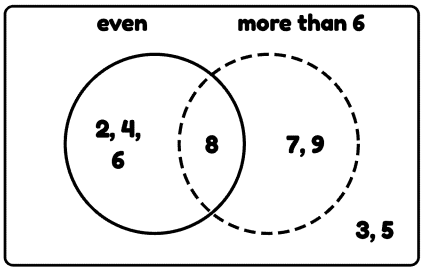
| | milk | juice | tea |
|---|---|---|---|
| Ravi | ✗ | ✗ | ✓ |
| Jenny | ✗ | ✓ | ✗ |
| Wei | ✓ | ✗ | ✗ |

1. Ravi picks tea. Tick it. Cross out the rest of that row and column.
2. Jenny does not pick milk. Cross out that box.
3. Jenny's last empty box is juice. Wei gets milk.

**Answer: Ravi: tea. Jenny: juice. Wei: milk.**

## Worked example 2

**Sort 2 to 9 into the loops. One rule is even. One rule is more than 6.**




1. Test each number against both rules.
2. 8 is even and more than 6. It goes in the middle.
3. 3 and 5 are neither. They go outside both loops.

**Answer: In both loops: 8. Outside both loops: 3 and 5.**

## How to use grids and loops

1. Read a clue. Tick or cross.
2. Tick a row's last empty box.
3. Test numbers against both rules.

## Odd and even

- Even: ends in 0, 2, 4, 6 or 8.
- Odd: ends in 1, 3, 5, 7 or 9.
- Counting in 4s: 4, 8, 12, 16 ...

## Your turn

Answer at the back of the book.

Try **Mei, Ben and Kumar each have 1 toy. The toys are a ball, a kite and a drum. Ben has the drum. Mei does not have the ball.**

Who has the ball? Mei Ben Kumar

| | ball | kite | drum |
|---|---|---|---|
| Mei | | | |
| Ben | | | |
| Kumar | | | |

## Start here

Do these first. Take your time.

**1** **Siti is older than Wei. Wei is older than Jamie.**

(a) The oldest is Siti Wei Jamie

(b) The youngest is Siti Wei Jamie

**2** **Ben is 3rd in a queue. Kumar stands just behind Ben. Aisha stands just in front of Ben.**

(a) Kumar is in place ☐.

(b) Aisha is in place ☐.

**3** **Ali, Mei, Ben and Siti stand in a line. Ali is 1st. Mei is just behind Ali. Ben is last.**

(a) Siti is in place ☐.

(b) The line from 1st to last:

_____

**4** **Ali is taller than Mei. Mei is taller than Ravi. Circle True or False.**

(a) Ravi is the shortest. True False

(b) Ali is taller than Ravi. True False

(c) Mei is the tallest. True False

**5** **Write each of these numbers in the right list: 4, 7, 10, 13, 16, 9.**

(a) Odd numbers: _____

(b) Even numbers: _____

## Start here

Keep going. Check your work.

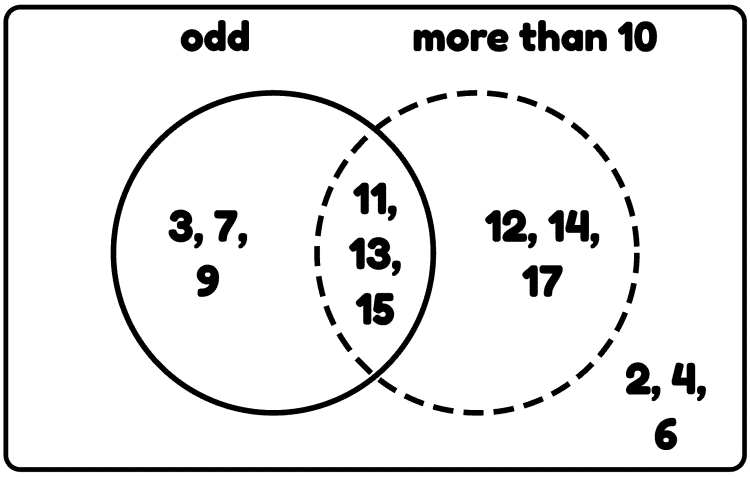
**6** **5 friends have 5, 6, 7, 8 and 9 sweets, all different. Mei has the most and Aisha has the fewest. Ravi has fewer sweets than Ben. Ben has fewer than Siti.**

(a) Ben has ☐ sweets.

(b) Siti has ☐ sweets.

Fewest to most: _____

**7** **One number is in the wrong place. Which number is it?** ☐




**8** **Count in 3s and count in 4s from 1 to 24.**

(a) Numbers said in both lists: _____

(b) Numbers said in 4s but not in 3s: ☐

Counting in 3s: _____

Counting in 4s: _____

## Start here

Nearly there. You can do it.

**9** **Ravi, Jenny and Wei each pick a different snack. Wei does not pick the bun or the pear. Jenny does not pick the apple. Ravi does not pick the pear.**

| | apple | bun | pear |
|---|---|---|---|
| Ravi | | | |
| Jenny | | | |
| Wei | | | |

(a) Ravi picks the apple bun pear

(b) Jenny picks the apple bun pear

(c) Wei picks the apple bun pear

**10** **Circle the odd one out.**

(a) 4 of these numbers are said counting in 4s. Which is not?

8 12 14 16 20

(b) 3 of these shapes have corners. Which does not?

triangle circle square

**11** **Find each number.**

(a) A number is between 10 and 20. It is odd. We say it counting in 5s. What is the number? ☐

(b) A number is even and between 20 and 30. We say it counting in 3s and in 4s. What is the number? ☐

## Think harder

Hints are upside down. Try first!

**1** **4 children each play a different instrument. Lily does not play the drum, flute or piano. Jason and Aisha do not play the piano. Jason does not play the bell or the drum.**

| | drum | flute | piano | bell |
|---|---|---|---|---|
| Lily | | | | |
| Jason | | | | |
| Kumar | | | | |
| Aisha | | | | |

(a) Kumar plays the drum flute piano bell

(b) Aisha plays the drum flute piano bell

**Hint.** Which row can you tick first?

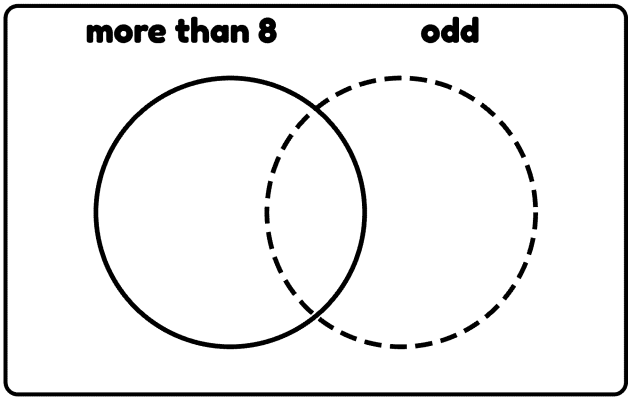
**2** **Imagine 2 loops. Loop A holds the numbers said counting in 2s. Loop B holds the numbers said counting in 5s. Use the numbers 1 to 20.**

(a) Numbers in both loops: ☐

(b) Numbers in no loop: ☐

**Hint.** Write the numbers said in 2s. Then the numbers said in 5s.

**3** **Sort the numbers 5 to 14 into the loops. Write each number where it belongs.**




(a) How many numbers are in both loops? ☐

(b) Which numbers are outside both loops?

_____

**Hint.** Sort the numbers one at a time. Test both rules.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 1: Truth and lies

**1** **One of 3 children ate the last biscuit. Exactly 1 child is not telling the truth. Who ate the biscuit?** Ali Ben Mei

| |
|---|
| Ali: "Ben ate it." |
| Ben: "I did not eat it." |
| Mei: "Ali ate it." |

**Hint.** Try each child as the one who ate it. Who is lying then?

**2** **Fill in the grid. Each row and each column has the numbers 1, 2, 3 and 4 once.**

| | | | |
|---|---|---|---|
| 2 | 1 | | |
| 4 | | | |
| | | | |
| 3 | | 4 | |

**Hint.** A row 3, 1, _, 4 needs a 2 in the gap. Find a line like that.

★ Beyond Level 1: Yes or no questions

**3** **Mei thinks of a secret number. Ali asks questions. Mei can only answer yes or no.**

**(a)** The number is from 1 to 4. After "Is it more than 2?" ☐ are left.

**(b)** The number is from 1 to 8. Ali always finds it with ☐ questions.

**(c)** The number is from 1 to 16. Ali always finds it with ☐ questions.

**Hint.** A good question splits the numbers into 2 equal groups.

## ✔ Check yourself

No hints. Try each one on your own.

**1** Siti runs faster than Ravi. Ravi runs faster than Jamie. Who is the slowest? Siti Ravi Jamie

Working: _____

**2** Use the numbers 1 to 10. How many are even and more than 5? ☐

Working: _____

**3** A number is between 20 and 30. It is odd. We say it counting in 5s. What is the number? ☐

Working: _____

**4** Ali is taller than Mei. Mei is taller than Ravi. Is Ravi taller than Ali?
Yes No

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

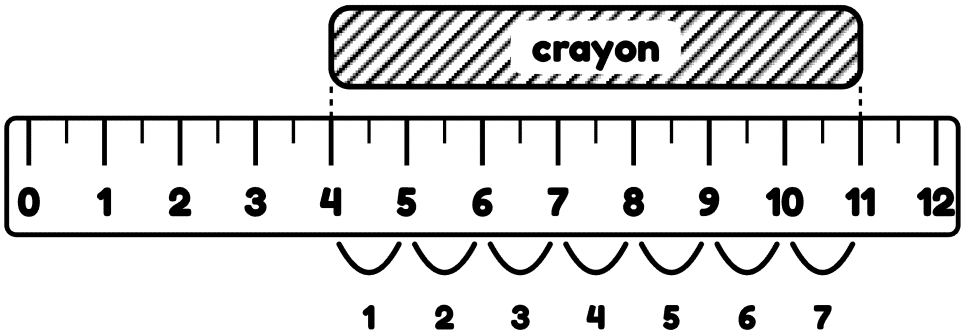
# Learn it: Length

**My goal. I can compare and measure lengths. I can use a ruler, also when the object does not start at 0.**

To measure, use equal units with no gaps. A cube, a clip and a centimetre (cm) are units. On a ruler, count the spaces, not the marks.

## Worked example

**A crayon lies on a ruler. It starts at the 4 cm mark and ends at the 11 cm mark. How long is the crayon?**




1. Find where the crayon starts: the 4 mark. It does not start at 0.
2. Find where it ends: the 11 mark.
3. Count the spaces from 4 to 11. Hop once for each space: 7 hops.
4. Check by subtracting: 11 - 4 = 7.

**Answer: The crayon is 7 cm long.**

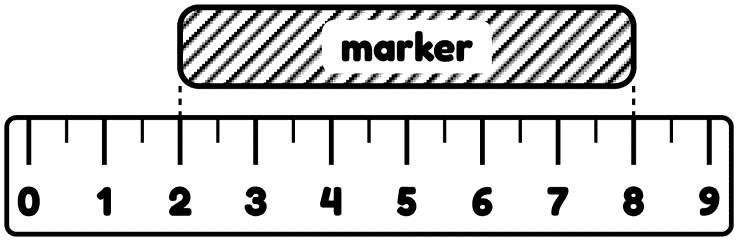
## How to measure with a ruler

1. Find the mark where the object starts.
2. Find the mark where it ends.
3. Count the spaces, or subtract: end mark - start mark.

## Your turn

Answer at the back of the book.

Try **A marker lies on the ruler. How long is the marker?** ☐ cm

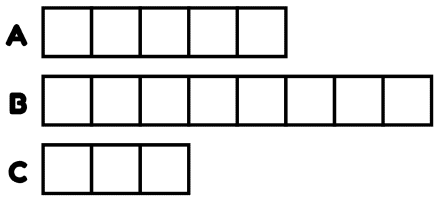


Name _____

## Start here

Do these first. Take your time.

**1** **Look at the 3 strips of cubes.**




(a) The longest strip is A B C

(b) The shortest strip is A B C

(c) Strip B is ☐ cubes longer than strip A.

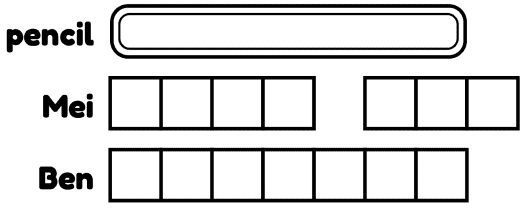
**2** **A tape is 13 cubes long. A strip is 8 cubes long. Fill in the blank. The tape is ☐ cubes longer.**

**3** **Arrange the lengths in order. Begin with the shortest.**
**14 cubes, 9 cubes, 12 cubes, 6 cubes**

_____

**4** **A rope is 15 cubes long. Ali cuts off 6 cubes. How long is the piece that is left? ☐ cubes**

**5** **Mei and Ben measure a pencil with cubes. Look at the pencil and the 2 rows.**




(a) Who measured fairly? Mei Ben

(b) What is wrong in the other row? a gap cubes overlap

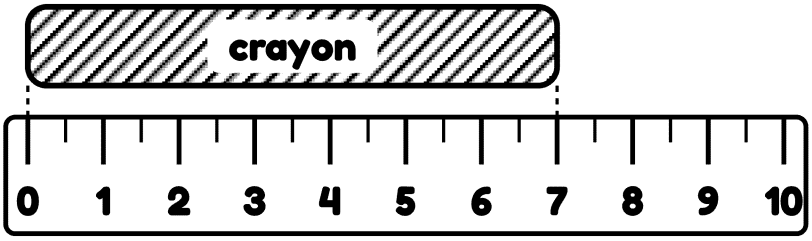
Name _____

## Start here

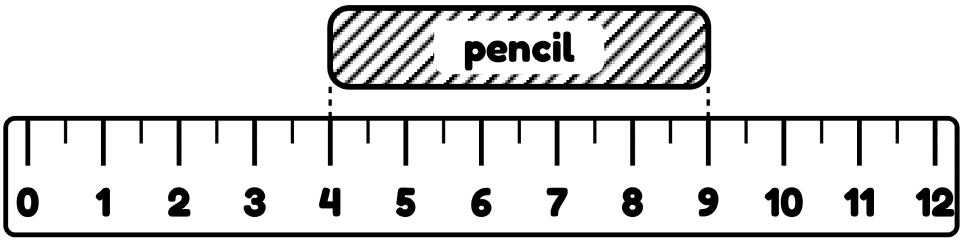
Keep going. Check your work.

**6** A tape is 8 cubes long. A strip is 5 cubes longer than the tape. How long is the strip? ☐ cubes

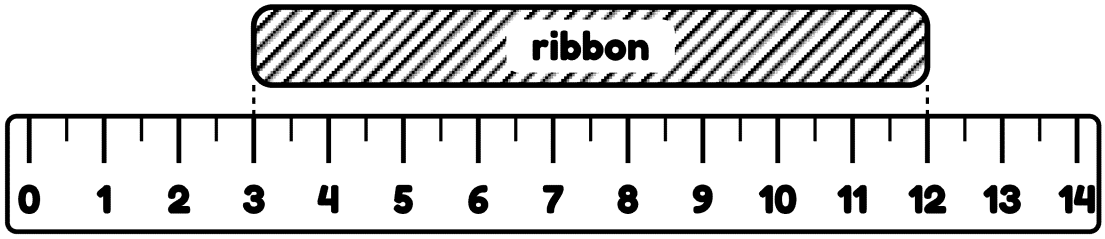
**7** Look at the ruler. How long is the crayon? ☐ cm




**8** Look at the ruler. How long is the pencil? ☐ cm




**9** Look at the ruler. How long is the ribbon? ☐ cm



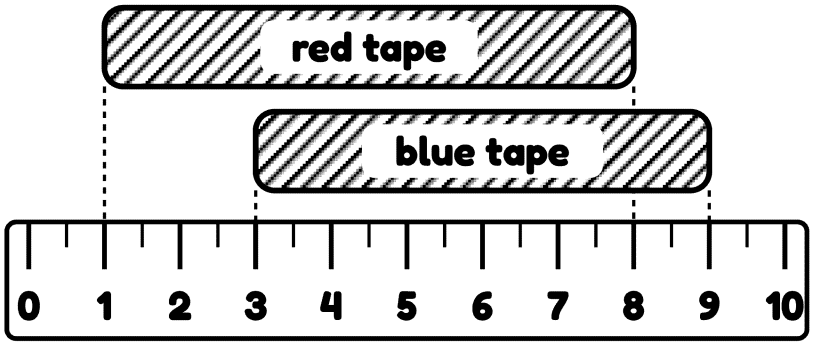

**10** A leaf lies between the 9 cm and 15 cm marks of a ruler. Write its length.

☐ cm

## Start here

Nearly there. You can do it.

**11** **Look at the ruler.**



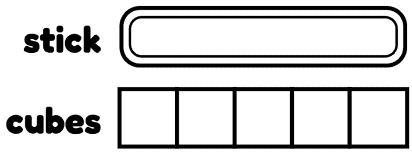

(a) The red tape is ☐ cm long.

(b) The blue tape is ☐ cm long.

(c) The red blue tape is longer.

**12** **A straw is 6 cm long. Its start is at the 4 cm mark of a ruler. Where is its end? The ☐ cm mark**

**13** **1 stick is as long as 5 cubes. Fill in the blanks.**




(a) 3 sticks are as long as ☐ cubes.

(b) 20 cubes are as long as ☐ sticks.

(c) 2 sticks and 3 cubes are as long as ☐ cubes.

**14** **A table is 6 pencils long. The same table is 9 rubbers long. Which is longer, a pencil or a rubber?** a pencil a rubber

## Think harder

Hints are upside down. Try first!

**1** **A ribbon is 24 cubes long. Ravi cuts it into 4 pieces of equal length.**

(a) Each piece is ☐ cubes long.

(b) Ravi makes ☐ cuts.

Draw the ribbon: _____

**Hint.** Draw the ribbon as one bar. Mark the 4 pieces.

**2** **A clip is the same length as 2 cubes. A pen is 5 clips long. A bag is 2 pens long.**

(a) The pen is ☐ cubes long.

(b) The bag is ☐ cubes long.

Pen: _____

Bag: _____

**Hint.** Change clips into cubes first.

**3** **A pencil starts at mark 3 of a ruler. It ends at mark 10. A crayon is 2 cm longer. It starts at mark 2.**

(a) The pencil is ☐ cm long.

(b) The crayon ends at the ☐ cm mark.

Pencil: _____

Crayon: _____

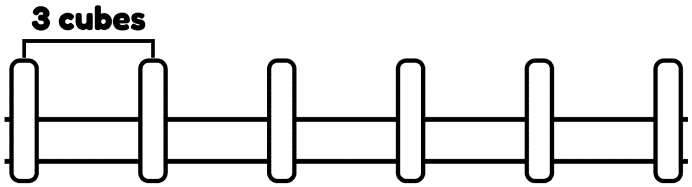
**Hint.** Find the pencil's length first. Then the crayon's.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 1: Posts and gaps

**1** **A fence has 6 posts in a row. Each gap between 2 posts is 3 cubes wide. Ignore how wide a post is.**




(a) The fence has ☐ gaps.

(b) From the first post to the last post is ☐ cubes.

(c) A fence with 10 posts has ☐ gaps.

**Hint.** Count the gaps, not the posts. How many gaps for 2 posts?

**2** **1 rod is as long as 4 cubes. Ali's rope is 3 rods and 2 cubes long. Mei's rope is 14 cubes long.**

(a) Ali's rope is ☐ cubes long.

(b) Which rope is longer? Ali's Mei's They are the same

Ali's rope: _____

**Hint.** Change the rods into cubes first.

**3** **A rope is 14 cubes long. Ben cuts it into pieces of 3 cubes or 4 cubes. Nothing is left over. How many pieces of each?**

☐ pieces of 4 cubes and ☐ pieces of 3 cubes.

**Hint.** Try 1 piece of 4 cubes. What is left?

 Name _____

## ✔ Check yourself

No hints. Try each one on your own.

**1** **A stick lies from 5 cm to 13 cm. How long is it?** ☐ **cm**

Working: _____

**2** **Ravi's string is 14 cubes. Lily's is 9. Ravi's is** ☐ **cubes longer.**

Working: _____

**3** **1 bar is as long as 3 cubes. 5 bars are as long as** ☐ **cubes.**

Working: _____

**4** **Tape A is 8 cubes. B is 3 cubes longer than A. C is 2 cubes shorter than A.**

**(a)** Which tape is longest? A B C

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

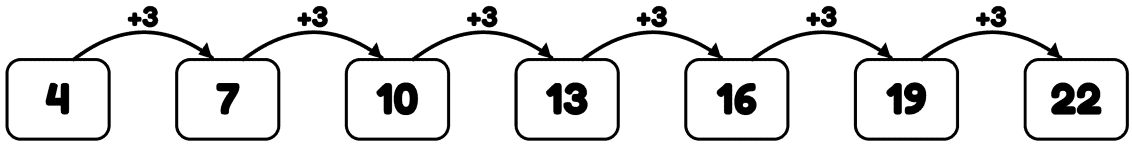
# Learn it: Mixed puzzles

**My goal.** I can pick a way to start a puzzle: draw it, list it or make a table. I can check and explain my answer.

Puzzles mix what you learned. Do not rush. Read it twice. Then draw it, list it or make a table.

**Worked example**

**Siti's plant is 4 cm tall. It grows 3 cm every week. How tall is the plant after 6 weeks?**




1. Say what we know. It starts at 4 cm and grows 3 cm a week.
2. List it. Write 4 first. Then add 3 for each week.
3. The list is 4, 7, 10, 13, 16, 19, 22. Week 6 is the last number.
4. Check: 6 weeks of 3 cm is 3 + 3 + 3 + 3 + 3 + 3 = 18. Then 4 + 18 = 22.

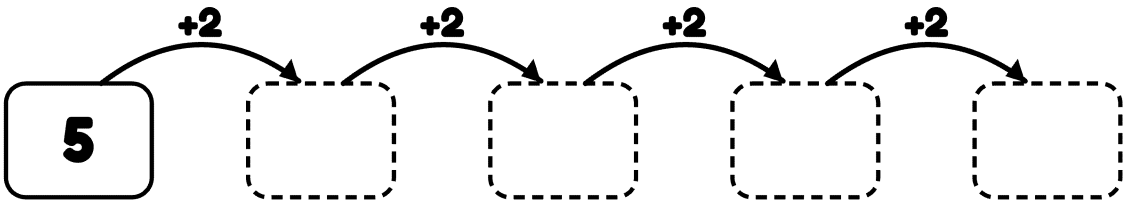
**Answer: After 6 weeks the plant is 22 cm tall.**

**How to start a puzzle**

1. Read it twice. What do I know? What must I find?
2. Draw it, list it or make a table.
3. Try. Then check your answer.
4. Say how you know.

**Your turn** Answer at the back of the book.

Try **A plant is 5 cm tall. It grows 2 cm every week. Write its height after each week in the tiles.**



## Start here

Do these first. Take your time.

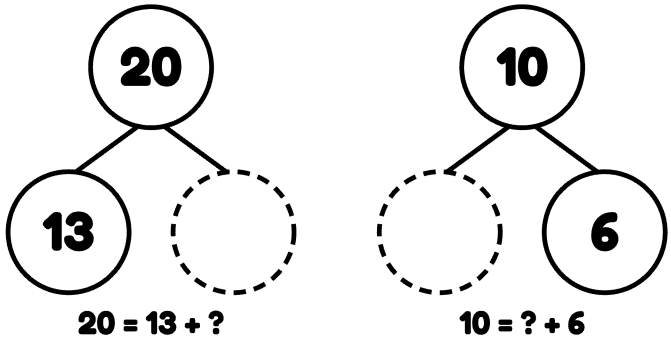
**1** **Find the rule. Write the missing number in the tile.**

| 7 | 12 | 17 | | 27 |
|---|---|---|---|---|

**2** **This pattern adds the same number each time. Write the missing numbers in the tiles.**

| 3 | 7 | | 15 | |
|---|---|---|---|---|

**3** **Fill in the missing numbers in the bonds.**




**4** **Fill in the missing numbers. Both sides must be equal.**

(a) 7 + 6 = 8 + ☐

(b) 9 + 5 = 7 + ☐

Working: _____

Name _____

## Start here

Keep going. Check your work.

**5** **Ravi has 70¢. He buys a drink for 45¢. How much money is left?** ☐ ¢

Working: _____

**6** **Look at the graph. Fill in the blanks.**

**Cookies sold**

| | |
|---|---|
| Monday | ●●●●●● |
| Tuesday | ●●●●●●●●● |
| Wednesday | ●●●● |
| Thursday | ●●●●●●● |

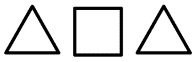
Key: ● = 1 cookie

(a) Tuesday had ☐ more cookies than Wednesday.

(b) Monday and Thursday had ☐ cookies altogether.

**7** **Count the sides of all the shapes. How many sides are there altogether?**

☐ **sides**



**8** **Lily is older than Ray. Jamie is younger than Ray. Amelia is older than Lily. Write the names, oldest first.**

_____

Draw it: _____

**9** **Circle the even numbers.** **13** **16** **18** **21** **24**

## Start here

Nearly there. You can do it.

**10** Make 40¢ with exactly 3 coins. Use 5¢, 10¢, 20¢ or 50¢ coins.

☐ ¢ + ☐ ¢ + ☐ ¢

Try: _____

**11** Mei picks a secret number. She adds 7. She gets 15. What number did Mei pick? ☐

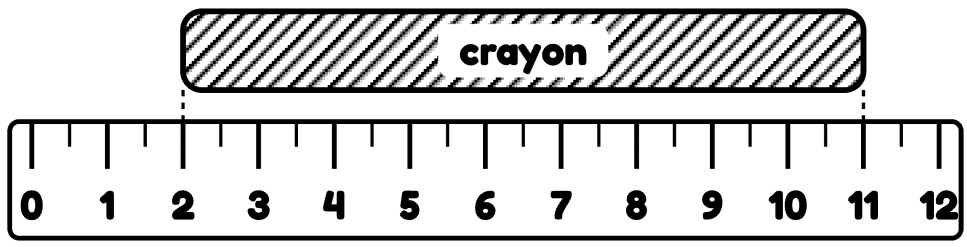
**12** Ben picks a secret number. He takes away 3. Then he adds 8. He gets 14. What number did Ben pick? ☐

Undo +8: _____

Undo -3: _____

**13** A ribbon is 13 cubes long. A string is 5 cubes shorter than the ribbon. How long is the string? ☐ cubes

**14** The ruler shows a crayon. What is the crayon's length? ☐ cm



## Think harder

Hints are upside down. Try first!

**1** **Mei saves 5¢ on day 1, 10¢ on day 2 and 15¢ on day 3. The pattern keeps going.**

(a) On day 4 she saves ☐ ¢.

(b) After 4 days she has saved ☐ ¢ altogether.

| Day | Saves | Total so far |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

**Hint.** Fill in the table. Add each new amount to the total so far.

**2** **Ravi puts 1 sticker on each side of 1 square and 2 triangles. Each sticker costs 5¢.**

(a) The 3 shapes have ☐ sides altogether.

(b) The stickers cost ☐ ¢ altogether.

Sides: _____

Cost: _____

**Hint.** Count the sides first. Then count in 5s.

**3** **Ali cuts 4 cubes off a rope, then 3 more cubes. He adds 2 cubes. The rope is now 11 cubes long. How long was it at first?** ☐ **cubes**

Undo +2: _____

Undo -3: _____

Undo -4: _____

**Hint.** Start at 11. Undo the last step first.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 1: Doubling

**1** **Mei folds a paper strip in half. 1 fold makes 2 equal parts. She folds again and again. Each new fold makes twice as many parts.**

(a) 3 folds make ☐ parts.

(b) 5 folds make ☐ parts.

(c) 16 parts need ☐ folds.

**Hint.** Write the number of parts after each fold.

**2** **Aisha, Ben and Mei have 24 stickers altogether. Ben has 4 more than Aisha. Mei has 2 fewer than Ben.**

(a) Aisha has ☐ stickers.

(b) Ben has ☐ stickers.

(c) Mei has ☐ stickers.

Guess: _____

**Hint.** Guess Aisha's number. Does the total come to 24?

**3** **Ravi has 3 triangles and some squares. He counts 17 sides altogether.**

(a) The 3 triangles have ☐ sides.

(b) The squares have ☐ sides.

(c) Ravi has ☐ squares.

**Hint.** Count the sides of the 3 triangles first.

## ✔ Check yourself

No hints. Try each one on your own.

**1** What are the next 2 numbers? 8, 13, 18, 23, ☐, ☐

Working: _____

**2** Fill in the missing number. 20 = 8 + ☐

Working: _____

**3** A pen costs 35¢. Ali pays with a 50¢ coin. How much change? ☐ ¢

Working: _____

**4** Wei is taller than Siti. Ben is shorter than Siti. Who is the tallest?

Wei Siti Ben

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a grown-up. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# My progress

Tick each box when you finish. Write your Check yourself score.

| Unit | Topic | Learn it | Start here | Think harder | Go further | Score | Date |
|---|---|---|---|---|---|---|---|
| 1 | Hops and rules | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 2 | Make 10 and 20 | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 3 | Read and draw graphs | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 4 | Patterns that go on | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 5 | Back to the start | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 6 | Pay the exact amount | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 7 | Shape detective | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 8 | Clues and sorting | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 9 | How long is it? | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 10 | Mix and solve | ☐ | ☐ | ☐ | ☐ | / 4 | |

## Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____




# Certificate of Completion

## Well done,

_____

**You finished the Math Inquirers Extension Workbook, Level 1.**

_____ _____

Date Signed by a grown-up

# Answers

For grown-ups. Please let your child try each question first, then read the short reason together.

Want to keep the answers out of sight while your child works? Print the answer pages on their own and keep them in an envelope, or cover them with a folded sheet of paper.

# Unit 1 · Hops and rules

**Your turn** 8, 13, 18, 23
**Why:** Add 5 each time: 3 + 5 = 8, 8 + 5 = 13, 13 + 5 = 18, 18 + 5 = 23.

## START HERE

**1** 12, 16
**Why:** Each hop is 2. 10 and 2 more is 12. 14 and 2 more is 16.

**2** 25, 35
**Why:** Each hop is 5. 20 and 5 more is 25. 30 and 5 more is 35.

**3** 37, 47
**Why:** Each hop is 10. The tens digit goes up by 1 and the ones digit stays 7.

**4** **(a)** 21, 25 **(b)** add 4
**Why:** From 5 to 9 is 4 more, and from 9 to 13 is 4 more. So the rule is add 4: 17 + 4 = 21, 21 + 4 = 25.

**5** **(a)** 9, 6 **(b)** take away 3
**Why:** The numbers go down by 3 each time: 12 - 3 = 9 and 9 - 3 = 6.

**6** add 4
**Why:** 7 to 11 is 4 more. 11 to 15 is 4 more. 15 to 19 is 4 more.

**7** **(a)** Down **(b)** take away 4 **(c)** 14
**Why:** 30 is more than 26, so the numbers go down. 30 - 26 = 4, so take away 4. 18 - 4 = 14.

**8** **(a)** 15 stickers **(b)** 18 stickers
**Why:** 3, 6, 9, 12, 15, 18. The 4th box has 12, the 5th has 15 and the 6th has 18.

**9** **(a)** 7 blocks **(b)** 2
**Why:** Step 1 has 1 block, Step 2 has 3 and Step 3 has 5. Each step adds 2 blocks, so Step 4 has 5 + 2 = 7.

**10** 17 is wrong. The right number is 18.
**Why:** The pattern adds 3 each time: 9, 12, 15, 18, 21, 24. So 17 should be 18.

**11** 26, 22
**Why:** Counting back means taking away 2 each time: 28 - 2 = 26 and 24 - 2 = 22.

**12** 25¢
**Why:** Each day he puts in 5¢ more: Thursday 20¢, Friday 25¢.

**13** 3, 7, 11, 15, 19, 23, 27, 31
**Why:** Start on 3 and count on 4 each time: 3, 7, 11, 15, 19, 23, 27, 31.

**14** 15, 10, 5, 0
**Why:** Counting back in 5s means taking away 5 each time: 20 - 5 = 15, 15 - 5 = 10, 10 - 5 = 5, 5 - 5 = 0.

## THINK HARDER

**1** **(a)** Yes **(b)** No
**Why:** 2, 7, 12, 17, 22, 27, 32, 37, 42. She lands on 37. She jumps from 37 over 40 to 42.

**2** **(a)** 20 and 40 **(b)** 20
**Why:** Ali says 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48. Mei says 10, 20, 30, 40, 50. They both say 20 and 40, and 40 - 20 = 20.

**3** **(a)** 3 hops **(b)** add 5 **(c)** 5
**Why:** From the 2nd to the 5th number is 3 hops (2nd to 3rd, 3rd to 4th, 4th to 5th). They make 25 - 10 = 15, so each hop is 15 / 3 = 5. Go back one hop from 10: the 1st number is 5.

### GO FURTHER

**1** **(a)** 10 dots **(b)** 15 dots **(c)** 28 dots
**Why:** Each triangle has one more row than the one before, and the new row has one more dot. So the extra dots are 2, 3, 4, 5, 6, 7: 1, 3, 6, 10, 15, 21, 28.

**2** 7, 22 and 37
**Why:** Mei says 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, 37, 40. Ali says 2, 7, 12, 17, 22, 27, 32, 37. The same numbers are 7, 22 and 37, and each is 15 more than the one before.

**3** 1, 2, 3, 6, 9, 18
**Why:** You need equal hops that make 18 from 2 to 20. Hops of 1 (18 of them), 2 (9), 3 (6), 6 (3), 9 (2) and 18 (1) all work.

### CHECK YOURSELF

**1** 30, 34
**Why:** The rule is add 4: 26 + 4 = 30 and 30 + 4 = 34.

**2** 20, 15; take away 5
**Why:** The numbers go down by 5 each time: 25 - 5 = 20 and 20 - 5 = 15.

**3** Yes
**Why:** Add 4 each time: 6, 10, 14, 18, 22, 26, 30. It lands on 30.

**4** 27 is wrong. The right number is 26.
**Why:** The pattern adds 5: 11, 16, 21, 26, 31, 36.

## Unit 2 · Make 10 and 20

**Your turn** 3
**Why:** 7 and 3 make 10, so 17 and 3 make 20. Check: 17 + 3 = 20.

### START HERE

**1** **(a)** 6 counters **(b)** 10 = 4 + 6
**Why:** The frame has 10 spaces. 4 are full, so 6 are empty. 4 + 6 = 10.

**2** **(a)** 2 **(b)** 7 **(c)** 5 **(d)** 9
**Why:** Think of the partner that makes 10: 8 and 2, 7 and 3, 5 and 5, 1 and 9.

**3** 4 marbles
**Why:** 6 and 4 make 10, so 4 marbles are in the bag.

**4** 9, 7, 4, 2
**Why:** 1 + 9 = 10, 3 + 7 = 10, 6 + 4 = 10, 8 + 2 = 10.

**5** 5 and 14
**Why:** First bond: 12 = 7 + 5. Second bond: the two parts 6 and 8 make 14.

| | |
|---|---|
| **6** | **(a)** 5 **(b)** 6 **(c)** 2 **(d)** 9<br>**Why:** 5 + 5 = 10, 4 + 6 = 10, 8 + 2 = 10, 1 + 9 = 10. So the partners to 20 are 5, 6, 2 and 9. |
| **7** | 3, 3, 3<br>**Why:** The ones digit is 7 each time. 7 + 3 = 10, so each number needs 3 more to reach the next ten. |
| **8** | 4 books<br>**Why:** 16 + 4 = 20, because 6 + 4 = 10. |
| **9** | **(a)** 4 **(b)** 5 **(c)** 8<br>**Why:** The sign = means both sides have the same value. 6 + 5 = 11, and 7 + 4 = 11. 8 + 2 = 10, and 5 + 5 = 10. 9 + 4 = 13, and 8 + 5 = 13. |
| **10** | **(a)** 7, 6, 5 **(b)** down<br>**Why:** 2 + 7, 3 + 6 and 4 + 5 all make 9. When one number goes up by 1, the other goes down by 1, so the total stays 9. |
| **11** | **(a)** True **(b)** False **(c)** True **(d)** False<br>**Why:** 8 + 3 = 11, not 10. 15 + 6 = 21, not 20. The other two are correct bonds. |
| **12** | 13 boys<br>**Why:** 7 + 3 = 10, so 7 + 13 = 20. There are 13 boys. |
| **13** | **(a)** 2 stickers **(b)** 12 stickers<br>**Why:** 8 + 2 = 10. From 10 to 20 is 10 more, so 8 + 12 = 20. |

## THINK HARDER

| | |
|---|---|
| **1** | 6 ways<br>**Why:** The ways are 1 and 13, 2 and 12, 3 and 11, 4 and 10, 5 and 9, 6 and 8. 7 and 7 is not allowed because the first bag must have fewer. |
| **2** | **(a)** ★ = 6 **(b)** ● = 7<br>**Why:** (a) 17 - 5 = 12, and 6 + 6 = 12, so ★ = 6. (b) Three equal numbers make 21: 7 + 7 + 7 = 21, so ● = 7. |
| **3** | 1 + 2 + 6, 1 + 3 + 5 and 2 + 3 + 4<br>**Why:** With 1 as the smallest: 1 + 2 + 6 and 1 + 3 + 5. With 2 as the smallest: 2 + 3 + 4. The smallest number cannot be 3 or more, because 3 + 4 + 5 = 12. |

## GO FURTHER

| | |
|---|---|
| **1** | **(a)** 4 stickers **(b)** 12 stickers<br>**Why:** 16 + 8 = 24 stickers. Shared equally, each child has 12. Ali goes from 16 to 12, so he gives 4. |
| **2** | **(a)** 11 sweets **(b)** 6 sweets<br>**Why:** 17 - 5 = 12. Shared equally, each jar gets 6. Jar A also has the 5 extra: 6 + 5 = 11. Check: 11 + 6 = 17. |
| **3** | 13, 23 and 33<br>**Why:** The numbers that end in 0 between 15 and 45 are 20, 30 and 40. Take away 7 from each: 13, 23 and 33. |

**4** **(a)** 3, 4, 5, 6, 7, 8, 9 **(b)** 3, 4, 8 and 9
**Why:** The pairs are 1+2=3, 1+3=4, 1+4=5, 1+5=6, 2+3=5, 2+4=6, 2+5=7, 3+4=7, 3+5=8, 4+5=9. The totals are 3 to 9. 5, 6 and 7 can each be made two ways. 3, 4, 8 and 9 only one way.

### CHECK YOURSELF

**1** 8
**Why:** 2 and 8 make 10.

**2** 14
**Why:** 6 and 4 make 10, so 6 and 14 make 20.

**3** 3
**Why:** 7 + 5 = 12. 9 and 3 make 12.

**4** 9 flowers
**Why:** 11 + 9 = 20, because 1 + 9 = 10.

## Unit 3 · Read and draw graphs

**Your turn** 3 shells
**Why:** Ben has 5 shells. Kumar has 2 shells. 5 - 2 = 3. Check: 2 + 3 = 5.

### START HERE

**1** **(a)** 5 children **(b)** Marbles **(c)** Skipping and tag
**Why:** Count the faces in each row. Catch has 5. Marbles has 3, the fewest. Skipping and tag both have 7.

**2** **(a)** 3 children **(b)** 26 children
**Why:** (a) "How many more" means subtract: 7 - 4 = 3. (b) Every child chose 1 game, so add every row: 7 + 7 + 5 + 4 + 3 = 26.

**3** **(a)** True **(b)** False
**Why:** (a) Catch has 5 and hopscotch has 4, so it is true. (b) Marbles has 3 faces, not 4: false.

**4** **(a)** 7 children **(b)** Zoo
**Why:** 9 + 6 + 8 = 23. From 23 to 30 is 7, so 7 children chose the farm. Zoo has 9 votes, the most.

**5** **(a)** Mei **(b)** 3 stickers **(c)** 4 stickers
**Why:** Mei has 9 stars, the most. Ben has 3. Mei and Siti: 9 - 5 = 4.

**6** **(a)** 9 stickers **(b)** 5 stickers
**Why:** (a) 6 + 3 = 9. (b) Ben had 3 and gets 2 more: 3 + 2 = 5.

**7** 13 faces
**Why:** Draw 7 faces for water, 4 for milk and 2 for tea. Count them all: 7 + 4 + 2 = 13.

**8** **(a)** Wednesday **(b)** 5 cookies **(c)** 3 cookies
**Why:** Wednesday has 8 cookies, the most. Thursday has 5. Friday 7 and Monday 4: 7 - 4 = 3.

**9** 8 books
**Why:** "3 more than Ali" means add: 5 + 3 = 8. Draw 8 books in Mei's row.

**10** 5 cakes
**Why:** The rest is 12 - 7 = 5. Draw 5 pictures in the Cakes row. Check: 7 + 5 = 12.

### THINK HARDER

**1** **(a)** 11 children **(b)** 5 children
**Why:** Chess: 5 faces make 10 and the half face makes 11. Singing: 3 faces make 6. 11 - 6 = 5.

**2** We cannot tell
**Why:** Only 5 children were asked. Other children in the school may choose other games. So we cannot tell.

**3** **(a)** 6 children and 10 children **(b)** Graph Q
**Why:** Graph P has 6 faces and each is 1 child: 6 children. Graph Q has 2 faces and each is 5 children: 5 + 5 = 10. Graph Q has more children, even with fewer pictures.

### GO FURTHER

**1** **(a)** 5 shells **(b)** 6 shells
**Why:** Lily: 7 - 2 = 5. Ravi and Lily have 7 + 5 = 12. The rest is 24 - 12 = 12. Jamie and Ray share it equally: 6 each. Check: 7 + 5 + 6 + 6 = 24.

**2** **(a)** 5 kites **(b)** 7 balls **(c)** 6 ropes
**Why:** The extra toys are 2 + 1 = 3. 18 - 3 = 15. Shared equally among 3 rows: 5 each. Kites 5, balls 5 + 2 = 7, ropes 5 + 1 = 6. Check: 5 + 7 + 6 = 18.

**3** **(a)** 12 stickers **(b)** 4 pictures
**Why:** (a) 6 pictures of 2: 2, 4, 6, 8, 10, 12. (b) Count in 3s: 3, 6, 9, 12. That is 4 pictures.

### CHECK YOURSELF

**1** 8 books
**Why:** Count the books in Kumar's row: 8.

**2** 5 books
**Why:** Kumar read 8 and Aisha read 3. 8 - 3 = 5.

**3** 11 books
**Why:** 8 + 3 = 11.

**4** Aisha
**Why:** Aisha read 3 books. Kumar read 8 books. 3 is fewer.

## Unit 4 · Patterns that go on

**Your turn** square
**Why:** The core has 4 shapes. Place 8 ends a core. Place 9 is a circle and place 10 is a square.

### START HERE

**1** triangle
**Why:** The core is triangle, circle. It repeats. The last shape is a circle, so a triangle comes next.

**2** heart
**Why:** The core is star, heart, heart. Place 7 is a star. Places 8 and 9 are hearts and place 10 is a star. The 9th shape is a heart.

**3** white
**Why:** The core is dark, dark, white: 3 beads. Count in 3s: 3, 6, 9, 12. Place 12 is the end of a core, so it is white.

**4** N, O
**Why:** The core is M, N, O. After M comes N and then O.

**5** clap
**Why:** The core is clap, clap, stomp: 3 actions. Count in 3s: 3, 6, 9. Place 9 is a stomp. Place 10 starts a new core, so it is a clap.

**6** circle
**Why:** The core is square, star, star, circle: 4 shapes. Count in 4s: 4, 8, 12, 16. Place 16 ends a core, so it is a circle.

**7** **(a)** square **(b)** square **(c)** triangle
**Why:** The core has 3 shapes. Places 9 and 12 end a core (3, 6, 9, 12), so they are squares. Place 13 starts a new core: a triangle.

**8** Places 11, 14 and 17
**Why:** A circle comes once in every core of 3 shapes. So the circles are 3 places apart: 2, 5, 8, 11, 14, 17.

**9** **(a)** True **(b)** False **(c)** True
**Why:** Places 4, 8, 12 end a core, and places 1, 5, 9, 13 start a core. (a) 4 and 8 both end a core: true. (b) 10 is the 2nd shape of a core: false. (c) 13 starts a core: true.

**10** 8 circles
**Why:** The core has 3 shapes and 2 are circles. 12 shapes make 4 cores. Count the circles in 2s: 2, 4, 6, 8.

**11** **(a)** 11 beads **(b)** Day 7
**Why:** Day 1: 3. Then add 2 each day: 3, 5, 7, 9, 11, 13, 15. Day 5 is 11 and 15 is day 7.

**12** **(a)** 11 blocks **(b)** 17 blocks **(c)** Tower 8
**Why:** Each tower has 3 more blocks than the one before: 2, 5, 8, 11, 14, 17, 20, 23. Tower 4 has 11 and Tower 6 has 17. 23 is the 8th number, so it is Tower 8.

**13** **(a)** 5 straws **(b)** 7 straws **(c)** 9 straws
**Why:** 1 triangle uses 3 straws. Each new triangle shares a side, so it needs only 2 more straws: 3, 5, 7, 9.

## THINK HARDER

**1** **(a)** star **(b)** 12 stars **(c)** Place 8
**Why:** (a) The core has 3 shapes and 15 ends a core, so it is a star. (b) 18 shapes make 6 cores with 2 stars each: 12. (c) The stars are in places 2, 3, 5, 6, 8. The 5th star is in place 8.

**2** **(a)** 11 triangles **(b)** 1 straw
**Why:** The numbers of straws are 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23. 11 triangles use 23 straws. 12 triangles need 25, which is too many. 24 - 23 = 1 straw is left.

**3** **(a)** 20 stickers **(b)** Day 10
**Why:** Add 3 each day: 5, 8, 11, 14, 17, 20, 23, 26, 29, 32. Day 6 is 20. 32 is the 10th number, so it is day 10.

### GO FURTHER

**1** **(a)** 16 and 22 **(b)** 29
**Why:** The jumps are 1, 2, 3, 4 and then 5, 6, 7. So 11 + 5 = 16 and 16 + 6 = 22. Then 22 + 7 = 29. The numbers are 1, 2, 4, 7, 11, 16, 22, 29. The 8th is 29.

**2** **(a)** 21 straws **(b)** 15 straws **(c)** 6 straws
**Why:** Ravi: 3, 6, 9, 12, 15, 18, 21. Mei: 3, 5, 7, 9, 11, 13, 15. The gap is 21 - 15 = 6 straws.

**3** **(a)** Place 6 **(b)** Place 12
**Why:** Stars are in even places. Cs are in places 3, 6, 9, 12, 15, 18, 21, 24. The places in both lists are 6, 12, 18 and 24.

### CHECK YOURSELF

**1** moon
**Why:** The core is sun, moon, moon: 3 words. Place 9 ends a core (3, 6, 9), so it is a moon.

**2** Q
**Why:** The core has 4 letters. Place 16 ends a core, so place 17 is P and place 18 is Q.

**3** 19 blocks
**Why:** Add 4 each time: 3, 7, 11, 15, 19. Tower 5 has 19 blocks.

**4** 13 straws
**Why:** Count on in 2s: 3, 5, 7, 9, 11, 13. 6 triangles use 13 straws.

## Unit 5 · Back to the start

**Your turn** 9 beads
**Why:** She added 6, so undo it by taking away 6: 15 - 6 = 9. Check: 9 + 6 = 15.

### START HERE

**1** 8 counters
**Why:** Adding 5 made 13. Undo it by taking away 5: 13 - 5 = 8. Check: 8 + 5 = 13.

**2** **(a)** 5 **(b)** 8 **(c)** 14 **(d)** 18
**Why:** Undo the step. Undo add with take away: 13 - 8 = 5 and 15 - 7 = 8. Undo take away with add: 9 + 5 = 14 and 12 + 6 = 18.

**3** 13 biscuits
**Why:** She took away 4 and 9 are left. Undo it by adding 4: 9 + 4 = 13. Check: 13 - 4 = 9.

**4** 8
**Why:** The arrow adds 9 to reach 17. Go backwards: 17 - 9 = 8. Check: 8 + 9 = 17.

**5** 9 pencils
**Why:** Start at the end: 14. Undo buying 8: 14 - 8 = 6. Undo giving 3 away: 6 + 3 = 9. Check: 9 - 3 = 6 and 6 + 8 = 14.

**6** 18 in the middle, 11 at the start
**Why:** The last arrow takes away 3 to reach 15. Undo it: 15 + 3 = 18. The first arrow adds 7 to reach 18. Undo it: 18 - 7 = 11.

**7** **(a)** 4 **(b)** 13, 13, 19
**Why:** IN 6 gives OUT 10, so the machine adds 4. IN 9 gives 9 + 4 = 13. OUT 17 came from IN 17 - 4 = 13. IN 15 gives 15 + 4 = 19.

**8** **(a)** 9 **(b)** 13 **(c)** 7
**Why:** Part (a): 15 - 6 = 9. Part (b): 9 + 4 = 13. Part (c): the number taken away is 18 - 11 = 7.

**9** floor 9
**Why:** Undo going up 6: 11 - 6 = 5. Undo going down 4: 5 + 4 = 9. Check: 9 - 4 = 5 and 5 + 6 = 11.

**10** 8 children
**Why:** Kumar is the 6th child, so 6 children are at the front, counting him. 2 more are behind him: 6 + 2 = 8 children.

**11** 7 children
**Why:** 5 children are at the front, and Aisha is the 5th. The rest are behind her: 12 - 5 = 7.

**12** **(a)** 9 **(b)** 15
**Why:** (a) Ali has more, so Siti has fewer: 15 - 6 = 9. (b) Ben has fewer, so Jenny has more: 9 + 6 = 15.

**13** **(a)** True **(b)** False **(c)** True
**Why:** (a) 9 + 4 = 13, 13 - 6 = 7: True. (b) 12 - 5 = 7, 7 + 3 = 10, not 9: False. (c) 8 + 7 = 15, 15 - 4 = 11: True.

**14** **(a)** No **(b)** 9
**Why:** To undo add 6, take away 6: 15 - 6 = 9. If his number were 21, adding 6 would give 27, not 15. So 21 is not his number.

## THINK HARDER

**1** **(a)** 17 **(b)** 9 **(c)** 14
**Why:** Undo the last step first: 13 + 4 = 17. Undo getting 8: 17 - 8 = 9. Undo giving 5 to Ali: 9 + 5 = 14. Check: 14 - 5 = 9, 9 + 8 = 17, 17 - 4 = 13.

**2** **(a)** 5 **(b)** 16 **(c)** 25
**Why:** Adding 9 and taking away 4 is the same as adding 5. IN 11 gives 11 + 5 = 16. For OUT 30, go backwards: 30 - 5 = 25.

**3** **(a)** 14 **(b)** 6
**Why:** Ali gave 4 away, so he had 4 more at first: 10 + 4 = 14. Mei got 4, so she had 4 fewer at first: 10 - 4 = 6. Check: 14 - 4 = 10 and 6 + 4 = 10.

## GO FURTHER

**1** **(a)** 12 **(b)** 7 **(c)** 8
**Why:** (b) Undo add 4: 18 - 4 = 14. 7 + 7 = 14, so her number is 7. (c) Undo take away 5: 11 + 5 = 16. 8 + 8 = 16, so her number is 8.

**2** **(a)** 9 **(b)** 9 **(c)** 2 **(d)** No
**Why:** 12 + 6 = 18, 18 - 9 = 9, and 12 - 9 = 3, 3 + 6 = 9. Both give 9. 5 + 6 = 11, 11 - 9 = 2. But you cannot take 9 from 5, so for 5 the order does matter.

**3** **(a)** day 5
**Why:** Start at the top and go backwards. On the last day the frog starts on step 8 and climbs 4 to reach 12. It starts each day 2 steps higher: day 1 starts on 0, day 2 on 2, day 3 on 4, day 4 on 6, day 5 on 8. So it gets out on day 5.

### CHECK YOURSELF

**1** 9
**Why:** Undo add 7: 16 - 7 = 9.

**2** 13
**Why:** Undo getting 9: 17 - 9 = 8. Undo giving away 5: 8 + 5 = 13.

**3** 15
**Why:** Undo take away 6 with add 6: 9 + 6 = 15.

**4** 15
**Why:** Undo take away 7 with add 7: 8 + 7 = 15. Check: 15 - 7 = 8.

## Unit 6 · Pay the exact amount

**Your turn** 50¢ + 20¢ + 10¢
**Why:** Take 50¢ first. 30¢ is left. 20¢ + 10¢ = 30¢. Check: 50¢ + 20¢ + 10¢ = 80¢.

### START HERE

**1** 70¢
**Why:** 50¢ and 10¢ make 60¢. Another 10¢ makes 70¢.

**2** $19
**Why:** $10 and $5 make $15. $2 and $2 make $4. $15 + $4 = $19.

**3** **(a)** 5 **(b)** 4 **(c)** 5 **(d)** 2
**Why:** (a) 10¢ five times is 50¢. (b) 5¢ four times is 20¢. (c) $1 is worth 100¢, and 20¢ five times is 100¢. (d) 50¢ and 50¢ make 100¢, which is $1.

**4** **(a)** 20¢ + 5¢ **(b)** 50¢ + 10¢ **(c)** 20¢ + 10¢
**Why:** (a) 20¢ + 5¢ = 25¢. (b) 50¢ + 10¢ = 60¢. (c) 20¢ + 10¢ = 30¢. Try the largest coin that fits first.

**5** 20¢
**Why:** 20¢ + 20¢ = 40¢. Two 10¢ coins make 20¢. Two 50¢ coins make 100¢.

**6** **(a)** 40 **(b)** ruler **(c)** 30
**Why:** (a) 15¢ + 25¢ = 40¢. (b) 45¢ is more than 30¢, so the ruler costs more. (c) 45¢ - 15¢ = 30¢.

**7** 20¢
**Why:** Change is what is left: 50¢ - 30¢ = 20¢. Count on to check: 30¢ and 20¢ more make 50¢.

**8** Kite and Puzzle
**Why:** $4 + $8 = $12. Check the others: $7 + $4 = $11, $7 + $8 = $15 and $11 + $4 = $15, so no other pair makes $12.

**9** $5
**Why:** She needs the difference: $13 - $8 = $5. Count on: $8 and $5 more make $13.

**10** **(a)** B **(b)** 5
**Why:** Purse A: 20¢ + 20¢ + 5¢ + 5¢ = 50¢. Purse B: 50¢ + 5¢ = 55¢. B has 55¢ - 50¢ = 5¢ more.

**11** **(a)** Kite **(b)** Ball
**Why:** $6 + $9 = $15 and $7 + $8 = $15. The Drum needs $3 more, and no toy costs $3.

**12** **(a)** True **(b)** False **(c)** True **(d)** False
**Why:** (a) 20 + 20 + 10 = 50: True. (b) three 10¢ coins make 30¢: False. (c) ten 10¢ coins make 100¢, which is $1: True. (d) four 5¢ coins make 20¢: False.

**13** **(a)** 2 **(b)** 3
**Why:** (a) 20¢ + 20¢ = 40¢. (b) 50¢ + 20¢ + 20¢ = 90¢. Start with the largest coin each time.

**14** 50¢, 10¢ and 5¢
**Why:** 50¢ + 10¢ + 5¢ = 65¢. Taking a 20¢ coin first leaves 45¢, which these coins cannot make.

## THINK HARDER

**1** 50¢ + 5¢ + 5¢ + 5¢ and 20¢ + 20¢ + 20¢ + 5¢
**Why:** With 50¢, 15¢ is left, made by 5¢ + 5¢ + 5¢. Without 50¢, try 20¢ + 20¢ + 20¢ = 60¢, then 5¢ makes 65¢. A 10¢ coin makes no more sets.

**2** **(a)** 4 **(b)** 3
**Why:** Try 4 coins of 5¢ and 3 coins of 10¢. 4 x 5¢ = 20¢ and 3 x 10¢ = 30¢. 20¢ + 30¢ = 50¢, and 4 + 3 = 7 coins.

**3** **(a)** 2 coins of 20¢ and 0 of 10¢; 1 and 2; 0 and 4 **(b)** 3
**Why:** Start with the most 20¢ coins. Two 20¢ coins make 40¢. One 20¢ and two 10¢ coins make 40¢. No 20¢ needs four 10¢ coins. That is 3 ways.

## GO FURTHER

**1** **(a)** 40 **(b)** 20 **(c)** No
**Why:** (a) 40¢ + 40¢ = 80¢. (b) 20¢ four times is 80¢. (c) Each child would get 25¢. Coins of 20¢ and 10¢ only make 10¢, 20¢, 30¢ and so on, never 25¢.

**2** **(a)** 38 **(b)** 20
**Why:** He spends $50 - $12 = $38 altogether. The bag costs the rest: $38 - $18 = $20. Check: $20 + $18 = $38 and $50 - $38 = $12.

**3** **(a)** 6 **(b)** $10, $10, $10, $5, $2, $1
**Why:** Three $10 notes make $30. $8 is left: $5 + $2 + $1. That is 3 + 3 = 6 pieces. Four $10 notes are too much, and no 5 pieces make $38.

## CHECK YOURSELF

**1** 60¢
**Why:** 20 + 20 = 40, 40 + 10 = 50, 50 + 5 = 55, 55 + 5 = 60.

**2** 50¢ + 10¢ + 10¢
**Why:** 50¢ + 10¢ + 10¢ = 70¢. Without a 50¢ coin, 3 coins make 60¢ at most.

**3** 25¢
**Why:** 90¢ - 65¢ = 25¢. Count on: 65¢ and 25¢ more make 90¢.

**4** $1
**Why:** A $1 coin is worth 100¢. 100¢ is more than 85¢, so $1 is more.

## Unit 7 · Shape detective

**Your turn** 3
**Why:** A pentagon has 5 sides. Touch each pentagon once. There are 3 of them. The circle, the triangle and the square do not have 5 sides.

### START HERE

**1** **(a)** pentagon **(b)** hexagon **(c)** octagon **(d)** triangle
**Why:** Count the sides. A has 5 sides: pentagon. B has 6: hexagon. C has 8: octagon. D has 3: triangle.

**2** **(a)** 5 **(b)** 7 **(c)** 0 **(d)** 8
**Why:** (a) A pentagon has 5 sides and 5 corners. (b) The number of corners is the same as the number of sides: 7. (c) A circle has no straight sides: 0. (d) 8 corners means 8 sides.

**3** **(a)** 4 **(b)** 3 **(c)** 12
**Why:** Count one kind at a time. Circles: 4. Triangles: 3. Squares: 4. The hexagon is 1 more. 4 + 3 + 4 + 1 = 12.

**4** **(a)** pentagon **(b)** square **(c)** octagon
**Why:** (a) 5 corners: pentagon. (b) 4 equal sides: square. (c) 8 straight sides: octagon.

**5** 2, 4, 6 and 8
**Why:** Shapes 2, 4, 6 and 8 are squares and rectangles. They have 4 corners. The others have 3, 0, 6 or 5 corners.

**6** **(a)** 10 **(b)** 9 **(c)** A
**Why:** (a) 5 in each of 2 rows: 5 + 5 = 10. (b) 3 in each of 3 rows: 3 + 3 + 3 = 9. (c) 10 is more than 9, so Grid A.

**7** **(a)** rectangle **(b)** 4
**Why:** The cut goes from the middle of the top to the middle of the bottom. Each piece is long and thin with 4 square corners: a rectangle.

**8** **(a)** rectangle **(b)** square
**Why:** (a) 2 squares in a row make a long shape with 4 corners: a rectangle. (b) 2 rows of 2 squares make a shape with 4 equal sides: a square.

**9** **(a)** 2 **(b)** 6
**Why:** (a) The line cuts the square into 2 triangles. (b) Each square is cut into 2 triangles. 3 squares make 3 x 2 = 6 triangles. The lines are side by side, so there are no larger triangles.

### THINK HARDER

**1** **(a)** 4 **(b)** 3 **(c)** 10
**Why:** 1 square long: 4. 2 squares long: 3. 3 squares long: 2. 4 squares long: 1. Altogether 4 + 3 + 2 + 1 = 10.

**2** **(a)** 6 **(b)** 8
**Why:** There are 6 small triangles. The first 2 lines meet at the bottom and make a larger triangle with the top of the strip. The last 2 lines meet at the top and make another. 6 + 2 = 8 triangles.

**3** **(a)** hexagon **(b)** 8
**Why:** (a) 11 - 5 = 6 sides: a hexagon. (b) 3 + 4 = 7 corners are known. 15 - 7 = 8 corners.

### GO FURTHER

**1** **(a)** 4 **(b)** 2 **(c)** 3
**Why:** (a) A square folds across, up and down, and along both corner-to-corner lines: 4. (b) A rectangle folds across and up and down: 2. A corner-to-corner fold does not match. (c) A triangle with 3 equal sides folds from each corner to the middle of the opposite side: 3.

**2** **(a)** 3 **(b)** 2
**Why:** (a) 12 tiles: 12 by 1, 6 by 2 and 4 by 3. That is 3 rectangles. (b) 9 tiles: 9 by 1 and 3 by 3. That is 2 rectangles, and one is a square.

**3** **(a)** 4 **(b)** 14
**Why:** There are 5 + 5 = 10 small squares. (a) A square made of 4 small squares can sit in 4 places along the grid. (b) 10 + 4 = 14. No larger square fits, because the grid is only 2 high.

### CHECK YOURSELF

**1** pentagon
**Why:** A shape with 5 sides is a pentagon.

**2** 12
**Why:** A square has 4 corners. 4 + 4 + 4 = 12.

**3** 4
**Why:** Each square is cut into 2 triangles. 2 + 2 = 4. The 2 lines are side by side, so there is no larger triangle.

**4** 6
**Why:** A triangle has 3 sides. 3 + 3 = 6 sides. It is a hexagon.

## Unit 8 · Clues and sorting

**Your turn** Kumar
**Why:** Ben has the drum. Mei does not have the ball or the drum, so Mei has the kite. The ball is left for Kumar.

### START HERE

**1** **(a)** Siti **(b)** Jamie
**Why:** Siti is older than Wei, and Wei is older than Jamie. The order from oldest is Siti, Wei, Jamie.

**2** **(a)** 4 **(b)** 2
**Why:** Just behind 3rd is 4th, so Kumar is 4th. Just in front of 3rd is 2nd, so Aisha is 2nd.

**3** **(a)** 3 **(b)** Ali, Mei, Siti, Ben
**Why:** Ali is 1st and Mei is just behind, so Mei is 2nd. Ben is last, 4th. Siti has the place that is left: 3rd.

**4** **(a)** True **(b)** True **(c)** False
**Why:** The order from tallest is Ali, Mei, Ravi. (a) True: Ravi is last. (b) True: Ali is first and Ravi is last. (c) False: Ali is the tallest.

**5** **(a)** 7, 13 and 9 **(b)** 4, 10 and 16
**Why:** Look at the last digit. 7, 13 and 9 end in 7, 3 and 9: odd. 4, 10 and 16 end in 4, 0 and 6: even.

**6** **(a)** 7 **(b)** 8
**Why:** The line is Aisha, Ravi, Ben, Siti, Mei (fewest to most). So Aisha has 5, Ravi 6, Ben 7, Siti 8 and Mei 9.

**7** 17
**Why:** 17 is odd and more than 10, so it belongs in both loops. It is in the "more than 10" loop only.

**8** **(a)** 12 and 24 **(b)** 4
**Why:** In 3s: 3, 6, 9, 12, 15, 18, 21, 24. In 4s: 4, 8, 12, 16, 20, 24. Both: 12 and 24. The 4s list has 6 numbers, and 2 of them are also in the 3s list, so 6 - 2 = 4 are in 4s only.

**9** **(a)** bun **(b)** pear **(c)** apple
**Why:** Wei does not pick the bun or the pear, so Wei picks the apple. Ravi does not pick the pear or the apple, so Ravi picks the bun. Jenny gets the pear.

**10** **(a)** 14 **(b)** circle
**Why:** (a) Counting in 4s we say 4, 8, 12, 16, 20. We do not say 14. (b) A circle has no corners.

**11** **(a)** 15 **(b)** 24
**Why:** (a) Counting in 5s we say 15. It is odd. (b) The even numbers are 22, 24, 26 and 28. Only 24 is said in 3s and in 4s.

## THINK HARDER

**1** **(a)** piano **(b)** drum
**Why:** Lily has the bell. Nobody except Kumar can have the piano (Lily, Jason and Aisha cannot), so Kumar has the piano. Jason does not have the bell or the drum, so Jason has the flute. Aisha has the drum.

**2** **(a)** 2 **(b)** 8
**Why:** Said in 2s: 2, 4, 6 ... 20. Said in 5s: 5, 10, 15, 20. In both: 10 and 20. 5s only: 5 and 15. 2s only: 8 numbers, so outside both there are 20 - 2 - 2 - 8 = 8 numbers.

**3** **(a)** 3 **(b)** 6 and 8
**Why:** More than 8: 9, 10, 11, 12, 13, 14. Odd: 5, 7, 9, 11, 13. In both: 9, 11 and 13. Outside both: 6 and 8, which are not more than 8 and not odd.

## GO FURTHER

**1** Ali
**Why:** Test each child. If Ali ate it, only Ali is not telling the truth. If Ben ate it, Ben and Mei are both wrong. If Mei ate it, Ali and Mei are both wrong. Only Ali gives exactly 1 wrong child.

**2** Row 1: 2 1 3 4. Row 2: 4 3 1 2. Row 3: 1 4 2 3. Row 4: 3 2 4 1.
**Why:** Column 1 has 2, 4 and 3, so its empty box is 1. Row 1 needs 3 and 4: column 3 already has a 4, so row 1 is 2 1 3 4. Row 4 needs 1 and 2: column 2 already has a 1, so row 4 is 3 2 4 1. Then fill each box that has only 1 number possible.

**3** **(a)** 2 **(b)** 3 **(c)** 4
**Why:** Each good question cuts the choices in half. 8 numbers become 4, then 2, then 1: 3 questions. 16 numbers become 8, 4, 2, 1: 4 questions.

### CHECK YOURSELF

**1** Jamie
**Why:** The order from fastest is Siti, Ravi, Jamie. Jamie is the slowest.

**2** 3
**Why:** The even numbers more than 5 are 6, 8 and 10. That is 3 numbers.

**3** 25
**Why:** Counting in 5s we say 25. It is between 20 and 30 and it is odd.

**4** No
**Why:** The order from tallest is Ali, Mei, Ravi. Ravi is the shortest, so he is not taller than Ali.

## Unit 9 · How long is it?

**Your turn** 6 cm
**Why:** The marker starts at 2 and ends at 8. Subtract: 8 - 2 = 6. Check by counting the spaces from 2 to 8: 6 hops.

### START HERE

**1** **(a)** B **(b)** C **(c)** 3
**Why:** Count the cubes: A has 5, B has 8, C has 3. B is longest and C is shortest. 8 - 5 = 3, so B is 3 cubes longer than A.

**2** 5 cubes
**Why:** "How many longer" asks for the gap. 13 - 8 = 5. Check: 8 + 5 = 13.

**3** 6, 9, 12, 14
**Why:** Find the smallest number first (6), then 9, 12 and 14. The shortest length has the fewest cubes.

**4** 9 cubes
**Why:** Cutting off takes away. 15 - 6 = 9. Check: 9 + 6 = 15.

**5** **(a)** Ben **(b)** a gap
**Why:** Ben's 7 cubes touch each other and fit the pencil exactly. Mei's cubes have a gap in the middle, so her row runs past the end of the pencil. Cubes must be end to end with no gaps.

**6** 13 cubes
**Why:** "Longer than" means add. 8 + 5 = 13. Check: 13 - 8 = 5.

**7** 7 cm
**Why:** The crayon starts at the 0 mark and ends at the 7 mark. It is 7 cm long.

**8** 5 cm
**Why:** The pencil starts at 4, not at 0. Count the spaces from 4 to 9: 5 spaces. Or subtract: 9 - 4 = 5.

**9** 9 cm
**Why:** Start at the 3 mark and end at the 12 mark. 12 - 3 = 9. Check: count 9 spaces.

**10** 6 cm
**Why:** Subtract the start from the end: 15 - 9 = 6. Check: from 9, hop 6 spaces to reach 15.

**11** **(a)** 7 **(b)** 6 **(c)** red
**Why:** Red goes from 1 to 8: 8 - 1 = 7 cm. Blue goes from 3 to 9: 9 - 3 = 6 cm. 7 is more than 6, so the red tape is longer.

**12** The 10 cm mark
**Why:** Start at 4 and add the length: 4 + 6 = 10. The straw ends at the 10 cm mark.

**13** **(a)** 15 **(b)** 4 **(c)** 13
**Why:** (a) 3 sticks are 5 + 5 + 5 = 15 cubes. (b) 20 cubes make 4 groups of 5, so 4 sticks. (c) 2 sticks are 10 cubes, and 3 more cubes make 13.

**14** a pencil
**Why:** The table is the same. The table needs fewer pencils, so each pencil is longer. A pencil is longer than a rubber. More pieces does not mean a longer piece.

## THINK HARDER

**1** **(a)** 6 **(b)** 3
**Why:** 4 equal pieces: 6 + 6 + 6 + 6 = 24, so each piece is 6 cubes. Cut between the pieces: 3 cuts make 4 pieces. The last piece needs no cut.

**2** **(a)** 10 **(b)** 20
**Why:** The pen is 5 clips, and each clip is 2 cubes: 2 + 2 + 2 + 2 + 2 = 10 cubes. The bag is 2 pens: 10 + 10 = 20 cubes.

**3** **(a)** 7 **(b)** 11
**Why:** The pencil is 10 - 3 = 7 cm long. The crayon is 2 cm longer, so it is 9 cm long. It starts at 2, so it ends at 2 + 9 = 11. Check: 11 - 2 = 9.

## GO FURTHER

**1** **(a)** 5 **(b)** 15 **(c)** 9
**Why:** (a) Posts and gaps alternate. The first post has no gap before it, so 6 posts have 5 gaps. (b) 5 gaps of 3 cubes: 3 + 3 + 3 + 3 + 3 = 15. (c) 10 posts have 10 - 1 = 9 gaps. There is always 1 gap fewer than posts.

**2** **(a)** 14 **(b)** They are the same
**Why:** Ali's rope: 3 rods are 4 + 4 + 4 = 12 cubes, and 2 more cubes make 14. Mei's rope is also 14 cubes. The ropes are the same length.

**3** Two 4-cube pieces and two 3-cube pieces
**Why:** One 4-cube piece leaves 10 cubes, and 3-cube pieces cannot fill 10. Two 4-cube pieces leave 6 cubes, and 3 + 3 = 6. So 4 + 4 + 3 + 3 = 14. Three 4-cube pieces leave 2 cubes, which a 3-cube piece cannot fill.

### CHECK YOURSELF

**1** 8 cm
**Why:** The stick starts at 5 and ends at 13. 13 - 5 = 8.

**2** 5 cubes
**Why:** 14 - 9 = 5. Check: 9 + 5 = 14.

**3** 15 cubes
**Why:** 5 bars: 3 + 3 + 3 + 3 + 3 = 15 cubes.

**4** **(a)** B
**Why:** B is 8 + 3 = 11 cubes. C is 8 - 2 = 6 cubes. A is 8 cubes. 11 is the greatest, so B is longest.

## Unit 10 · Mix and solve

**Your turn** 7, 9, 11, 13
**Why:** Add 2 each week: 5 + 2 = 7, 7 + 2 = 9, 9 + 2 = 11, 11 + 2 = 13. After 4 weeks the plant is 13 cm tall.

### START HERE

**1** 22
**Why:** Each tile is 5 more than the one before: 7, 12, 17. So the missing tile is 17 + 5 = 22. Check: 22 + 5 = 27.

**2** 11, 19
**Why:** From 3 to 7 is 4 more, so the rule is add 4. After 7 comes 11. After 11 comes 15. After 15 comes 19.

**3** 7 and 4
**Why:** 13 and 7 make 20, because 3 and 7 make 10. 6 and 4 make 10.

**4** **(a)** 5 **(b)** 7
**Why:** (a) 7 + 6 = 13. 8 and 5 make 13. (b) 9 + 5 = 14. 7 and 7 make 14.

**5** 25¢
**Why:** 70¢ - 45¢ = 25¢. Check: 45¢ + 25¢ = 70¢.

**6** **(a)** 5 **(b)** 13
**Why:** Tuesday has 9 cookies and Wednesday has 4. 9 - 4 = 5 more. Monday has 6 and Thursday has 7. 6 + 7 = 13.

**7** 10 sides
**Why:** A triangle has 3 sides and a square has 4 sides. 3 + 4 + 3 = 10.

**8** Amelia, Lily, Ray, Jamie
**Why:** Amelia is older than Lily, Lily is older than Ray, and Ray is older than Jamie. The order is Amelia, Lily, Ray, Jamie.

**9** 16, 18, 24
**Why:** An even number ends in 0, 2, 4, 6 or 8. 16, 18 and 24 are even. 13 and 21 end in 3 and 1: odd.

**10** 20¢ + 10¢ + 10¢
**Why:** 20¢ + 10¢ + 10¢ = 40¢. This is the only way: 50¢ is too big, and 5¢ coins would need 35¢ from 2 coins, which no 2 coins make.

**11** 8
**Why:** Go backwards. Undo adding 7 by taking away 7: 15 - 7 = 8. Check: 8 + 7 = 15.

**12** 9
**Why:** Go backwards. Undo the last step: 14 - 8 = 6. Undo the first step: 6 + 3 = 9. Check: 9 - 3 = 6 and 6 + 8 = 14.

**13** 8 cubes
**Why:** "Shorter than" means take away. 13 - 5 = 8. Check: 8 + 5 = 13.

**14** 9 cm
**Why:** The crayon starts at 2 and ends at 11. 11 - 2 = 9. Check: count 9 spaces.

## THINK HARDER

**1** **(a)** 20 **(b)** 50
**Why:** The amount grows by 5¢ each day: 5, 10, 15, 20. On day 4 she saves 20¢. Add each day to the total: 5 + 10 = 15, 15 + 15 = 30, 30 + 20 = 50.

**2** **(a)** 10 **(b)** 50
**Why:** A square has 4 sides and each triangle has 3: 4 + 3 + 3 = 10 sides. That is 10 stickers. Counting in 5s ten times gives 50, so the stickers cost 50¢.

**3** 16 cubes
**Why:** Go backwards. Undo adding 2: 11 - 2 = 9. Undo cutting off 3: 9 + 3 = 12. Undo cutting off 4: 12 + 4 = 16. Check: 16 - 4 = 12, 12 - 3 = 9, 9 + 2 = 11.

## GO FURTHER

**1** **(a)** 8 **(b)** 32 **(c)** 4
**Why:** Parts after each fold: 2, 4, 8, 16, 32. (a) 3 folds make 8 parts. (b) 5 folds make 32 parts. (c) 16 parts need 4 folds.

**2** **(a)** 6 **(b)** 10 **(c)** 8
**Why:** Try 6 for Aisha. Ben has 6 + 4 = 10. Mei has 10 - 2 = 8. Together 6 + 10 + 8 = 24. A smaller guess gives less than 24 and a greater guess gives more than 24.

**3** **(a)** 9 **(b)** 8 **(c)** 2
**Why:** A triangle has 3 sides: 3 + 3 + 3 = 9. The squares have 17 - 9 = 8 sides. A square has 4 sides, and 4 + 4 = 8, so there are 2 squares.

## CHECK YOURSELF

**1** 28, 33
**Why:** The rule is add 5. 23 + 5 = 28 and 28 + 5 = 33.

**2** 12
**Why:** 8 and 2 make 10, and 10 and 10 make 20. So 8 and 12 make 20.

**3** 15¢
**Why:** 50¢ - 35¢ = 15¢. Check: 35¢ + 15¢ = 50¢.

**4** Wei
**Why:** Wei is taller than Siti, and Siti is taller than Ben. So Wei is the tallest.